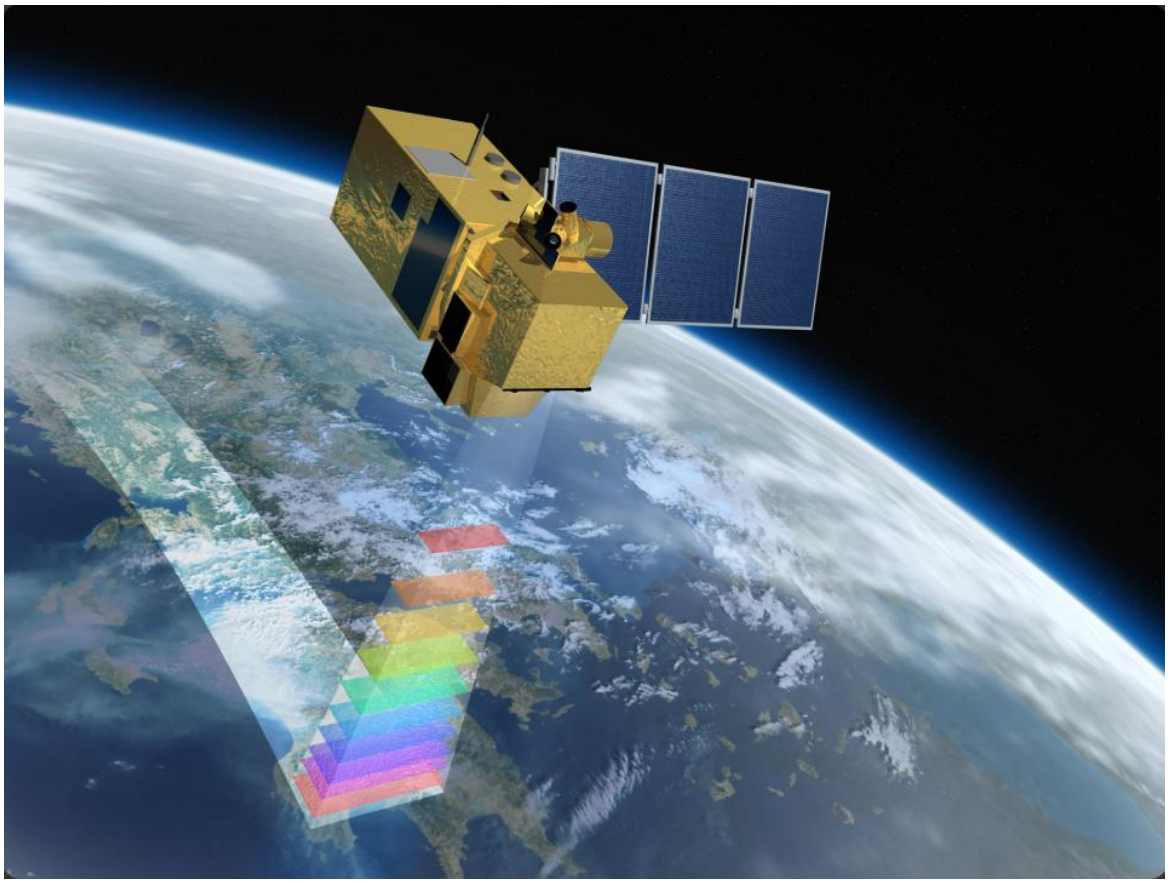
Pixel-Based Landslide Detection from Sentinel-2 and Terrain Data: A Statistical Comparison of Six Supervised Classifiers

Oct 21, 2026 · Nooshin Hadidian M.



Contents

Abstract **1**

1. Introduction **1**

2. Data **2**

2.1 Images and input channels 2

2.2 Labelled subset and data split 2

2.3 Exploratory data analysis 4

3. Methods **5**

3.1 Evaluation metric 5

3.2 Shared evaluation protocol 6

3.3 Classifiers 7

3.4 Diagnostics and explainability 7

3.5 Statistical comparison 7

4. Results **8**

4.1 Results of the individual classifiers 8

4.1.1 Logistic regression on the raw features 8

4.1.2 Logistic regression with pairwise interactions 10

4.1.3 Decision tree 12

4.1.4 Random forest 15

4.1.5 Histogram gradient boosting (HGB) 19

4.1.6 XGBoost 22

4.2 Overall performance 26

4.3 Statistical comparison 27

4.4 Landslide size and false alarms 28

4.5 Ranking quality and shared errors 30

4.6 Probability calibration and computing cost 31

4.7 What the classifiers rely on 32

5. Discussion **35**

5.1 Why the ensembles lead — and why they plateau 35

5.2 The physical reason for the remaining errors 35

5.3 Statistical versus practical significance 35

5.4 Which classifier to use 35

6. Limitations **36**

7. Conclusion and future work **36**

7.1 Conclusion 36

7.2 Future work 36

7.3 Further applications 37

References **38**

Appendix **39**

A. Size dependence of the random-forest advantage 39

B. Transfer from the outer to the test evaluation 39

C. Image-level outcomes 40

D. Secondary metrics among the ensembles 40

E. Normality check of the per-image differences 40

List of Figures

Figure 1. Example training patch 3
Figure 2. Class balance of the 3,799 labelled patches 3
Figure 3. Landslide coverage within the 2,231 patches that contain landslides 4
Figure 4. Median value of each input channel for non-landslide (blue) and landslide (orange) pixels 4
Figure 5. Spearman correlations between the 14 input channels on the development sample 5
Figure 6. Shared evaluation protocol 6
Figure 7. Logistic regression with spectral bands only, terrain only, and both 9
Figure 8. Standardised coefficients of the final logistic regression 9
Figure 9. Final logistic regression on the 760 test images 10
Figure 10. Original and interaction logistic regression on the five outer folds 11
Figure 11. Original versus interaction LR on the test set 11
Figure 12. Interaction logistic regression on three fixed test images 12
Figure 13. Final-development selection of the decision tree 12
Figure 14. The first three levels of the decision tree of outer fold 1 (orange = majority non-landslide, blue = majority landslide) 13
Figure 15. Decision tree feature importance across the five outer models 13
Figure 16. Final decision tree on the 760 test images: confusion matrix (counts and row percentages), ROC curve and precision–recall curve with the locked threshold 0.92 14
Figure 17. Reliability of the raw decision-tree scores (pooled out-of-fold predictions, 20 bins) 15
Figure 18. Effect of the number of trees on inner-validation F2 (outer fold 5, other settings fixed) 15
Figure 19. Random forest five-fold nested-CV results 16
Figure 20. False positives in the 1,254 landslide-free development images 16
Figure 21. Random forest permutation importance across the five outer models (F2 drop in percentage points; coloured points = folds, black marks = mean) 17
Figure 22. Final random forest on the 760 test images: confusion matrix as pixel counts and as percentages within each true class (threshold 0.17) 18
Figure 23. Random forest on three fixed test images 18
Figure 24. Random forest test performance by landslide coverage per image 19
Figure 25. HGB final development selection 19
Figure 26. HGB outer evaluation 20
Figure 27. Final HGB on the 760 test images: confusion matrix as pixel counts and as percentages within each reference class (threshold 0.13) 20
Figure 28. HGB calibration on the test set 21
Figure 29. HGB on six fixed test images 21
Figure 30. HGB channel permutation importance on the test set 22
Figure 31. XGBoost final development selection 22
Figure 32. XGBoost outer evaluation 23
Figure 33. XGBoost ROC and precision–recall curves on the 760 test images 23
Figure 34. F2, precision, recall, specificity and IoU of XGBoost across thresholds on the test set (diagnostic only) 24
Figure 35. XGBoost on six fixed test images 25
Figure 36. XGBoost channel permutation importance on the test set 26

Figure 37. Paired whole-image bootstrap of the F2 difference between each classifier and the random forest, with 95% confidence intervals 27
Figure 38. All pairwise F2 differences (row − column) on the outer (left) and test (right) evaluation 28
Figure 39. Bayesian bootstrap posteriors of the F2 difference for five key pairs 28
Figure 40. Pooled F2 of each classifier by landslide coverage of the image, for the outer (left) and test (centre) evaluation with 95% bootstrap intervals, and the share of falsely flagged pixels on images without landslides (right; solid = outer, hatched = test) 29
Figure 41. Distribution of per-image F2 on images with landslides (outer: 1,785; test: 446) 29
Figure 42. Image-level outcomes with 95% Bayesian credible intervals (Beta(1,1) prior) 30
Figure 43. Precision–recall curves of all classifiers on the test set with F2 iso-lines (dashed grey) 31
Figure 44. Model consensus on six fixed test images 31
Figure 45. Trade-offs on the test set 32
Figure 46. SHAP summary of the final XGBoost model on test pixels (balanced display sample of 4,000 landslide and 4,000 non-landslide pixels) 33
Figure 47. SHAP dependence of the four most important channels (B1, B4, elevation, B7) 34
Figure 48. SHAP explanation maps of three test images 34
Figure 49. Normal Q-Q plots of the per-image F2 differences (classifier − random forest) on the 446 test images with landslides 40

List of Tables

Table 1. Composition of the development and secondary test sets 3
Table 2. Selected configuration of each classifier 7
Table 3. Overall performance on the outer evaluation and the secondary test set 26
Table 4. Probability calibration and prediction time on the secondary test set 31
Table 5. Size dependence of the random-forest advantage (difference-in-differences) 39
Table 6. Change in F2 from the outer to the secondary test evaluation 39
Table 7. Image-level outcomes of each classifier 40

Abstract

Rapid and reliable landslide maps are essential after earthquakes and extreme rainfall, when emergency services must know quickly where slopes have failed. Optical satellite imagery can cover large areas within days, but mapping landslides automatically is difficult. Fresh landslide scars are small and rare, and other bright, bare surfaces such as riverbeds, roads and fields look very similar. This project examines how well supervised classifiers can detect landslides pixel by pixel from multispectral and terrain data. It also asks whether the differences between classifiers are real and practically meaningful.

Six classifiers were compared on the Landslide4Sense benchmark: logistic regression on the raw features, logistic regression with pairwise interaction features, a decision tree, a random forest, histogram gradient boosting (HGB) and XGBoost. The data consist of 128 × 128-pixel patches with 12 Sentinel-2 bands plus slope and elevation from ALOS PALSAR. Landslide pixels make up only 2.3% of the data, so the models were tuned and judged by the recall-weighted F2 score. All classifiers followed the same protocol: image-level nested cross-validation, thresholds selected on development data only, evaluation on every pixel, and a locked final model tested once on held-out images. Differences were tested with paired whole-image bootstrap tests, Wilcoxon signed-rank tests and a Bayesian analysis with a region of practical equivalence (±0.01 F2).

The classifiers form four clearly separated groups. The three ensembles lead, with F2 ≈ 0.77 on 3,039 development images and 0.765–0.767 on 760 held-out test images. They are followed by logistic regression with interaction features (0.743), the decision tree (0.723) and logistic regression on the raw features (0.612). The Bayesian analysis shows with posterior probability 1.00 that the three ensembles differ by less than 0.01 F2, so they are practically equivalent. They differ mainly in secondary properties: the random forest gives the best-calibrated probabilities, while XGBoost predicts 2.5 times faster at a far lower training cost.

All models share the same weaknesses. About one in two predicted landslide pixels is a false alarm, small landslides are often missed, and 8% of landslide pixels are missed by every model. SHAP explanations trace these errors to bright, bare surfaces that resemble fresh landslide scars in the red band. Pixel-based classification therefore appears to have reached a plateau. Adding spatial context, through deep learning or contextual features, is the most promising next step.

1. Introduction

Landslides are among the most frequent and destructive natural hazards in mountainous regions. Strong earthquakes and intense rainfall can trigger thousands of slope failures within hours. The debris blocks roads, cuts off villages, damages buildings and can dam rivers. Accurate maps of the affected slopes are needed to guide rescue teams, plan reconstruction and build landslide inventories for risk models. Field surveys, however, are slow and often impossible in steep or inaccessible terrain.

Earth observation satellites overcome these limits. Sentinel-2 provides free multispectral images at 10–20 m resolution and revisits every location every few days, so large areas can be covered soon after an event. The remaining challenge is to separate landslide pixels reliably from all other surfaces in the image.

Fresh landslides expose bare soil and rock that look bright and unvegetated in optical imagery. The same appearance also occurs on riverbeds, roads, construction sites and agricultural land, and landslides are rare compared with stable terrain. Landslide detection is therefore a difficult, strongly imbalanced classification problem in which accuracy alone is misleading: a model that predicts "no landslide" everywhere is 97.7% accurate on this dataset but useless.

This project applies the supervised classification methods covered in the course to this problem and compares them under one strict, shared evaluation protocol. It addresses four questions:

1. How well can pixel-based supervised classifiers detect landslides from multispectral and terrain data?
2. Which classifier performs best, and are the differences between classifiers statistically and practically meaningful?
3. Where and why do the classifiers fail — for which landslide sizes, which surfaces and which input channels?
4. Which classifier is the most sensible choice in practice once accuracy, probability quality and computing cost are considered together?

Six classifiers were compared: logistic regression on the raw features, logistic regression with pairwise interaction features, a decision tree, a random forest, histogram gradient boosting and XGBoost. Their differences were tested with frequentist and Bayesian methods, treating complete images rather than pixels as the unit of analysis.

2. Data

The project uses the Landslide4Sense benchmark dataset (Kaggle), created for the Landslide4Sense 2022 competition and described by Ghorbanzadeh et al. (2022). It contains labelled image patches from four regions affected by major landslide events: Iburi (Hokkaido, Japan), where an Mw 6.6 earthquake in September 2018 followed heavy rainfall from Typhoon Jebi; Kodagu (Karnataka, India), hit by exceptional monsoon rainfall in August 2018; the Rasuwa District in Nepal, affected by the Gorkha earthquake of April 2015; and western Taitung County in Taiwan, struck by Typhoon Morakot in August 2009.

2.1 Images and input channels

Each patch covers 128 × 128 pixels and stacks 14 co-registered layers, all resampled to a pixel size of about 10 m (dataset documentation):

- **B1–B12:** multispectral bands of the European Space Agency's Sentinel-2 satellites, covering visible light (B2 blue, B3 green, B4 red), the red edge (B5–B7), near-infrared (B8, B9) and shortwave infrared (B10–B12). The bands have native resolutions of 10, 20 or 60 m; B1 (coastal aerosol), B9 (water vapour) and B10 (cirrus) are 60 m atmospheric bands.
- **Slope and elevation:** terrain layers derived from a digital elevation model of the ALOS PALSAR mission (Japan Aerospace Exploration Agency).

Every pixel carries a binary label: 1 = landslide, 0 = no landslide. The stored values are scaled by the dataset providers (divided by per-channel mean values), so slope and elevation are used as relative values, not as degrees or metres.

2.2 Labelled subset and data split

Only the 3,799 training patches of the released dataset include reference masks; the official validation (245) and test (800) patches come without labels. All 3,799 labelled patches passed integrity checks (readable files, correct dimensions, finite values, binary masks) and contained no exact duplicates. They were split once into a **development set** (3,039 patches, 80%) and a **secondary test set** (760 patches, 20%), stratified by whether a patch contains any landslide (random seed 43).

Table 1. *Composition of the development and secondary test sets.*

Subset	Images	With landslides	Without landslides	Landslide pixels	Share of pixels
Development	3,039	1,785	1,254	1,177,409	2.36%
Secondary test	760	446	314	265,381	2.13%
Total	3,799	2,231	1,568	1,442,790	2.32%

The data are **strongly imbalanced**: about 42 non-landslide pixels for every landslide pixel. Landslides are also small. Among patches that contain landslides, the median landslide coverage is only 2.0% of the patch, and three quarters are covered by at most 5.2%. The released files carry no coordinates, so the exact location of each patch, and therefore the spatial independence of development and test patches, cannot be verified.

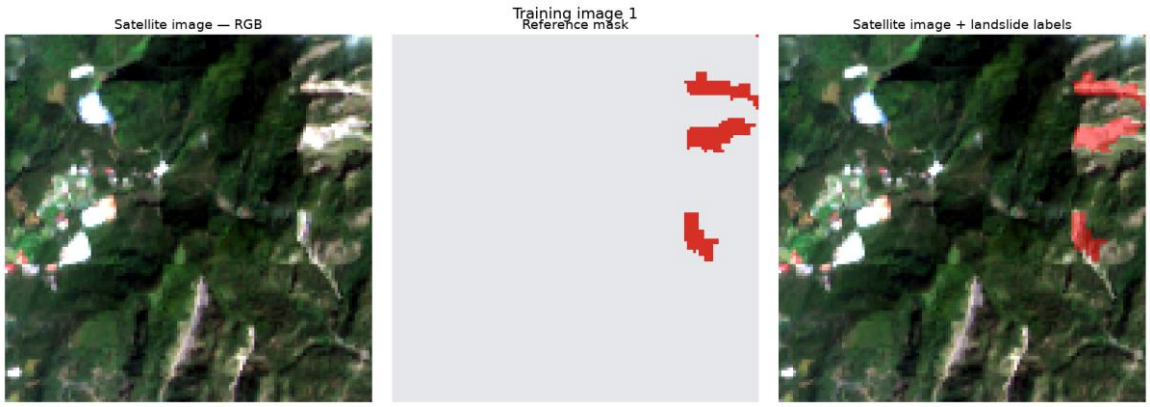


Figure 1. *Example training patch. Left: Sentinel-2 true-colour composite (B4, B3, B2). Centre: reference mask (red = landslide). Right: mask overlaid on the image. The landslides are small, bright, unvegetated scars on forested slopes.*

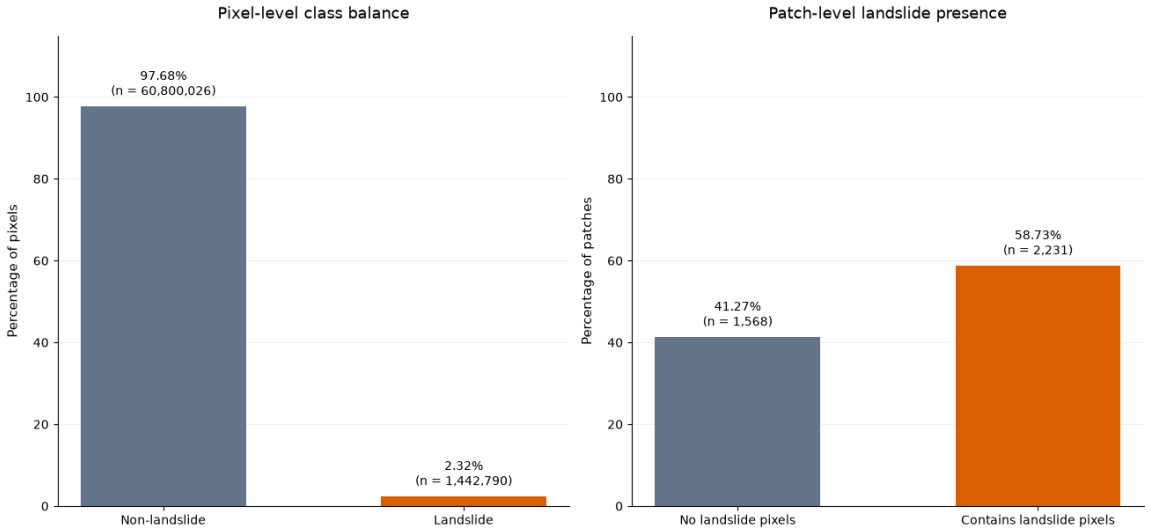


Figure 2. *Class balance of the 3,799 labelled patches. Left: only 2.32% of all pixels (1,442,790 of 62.2 million) are landslide pixels. Right: 58.7% of the patches (2,231) contain at least one landslide pixel, 41.3% (1,568) contain none.*

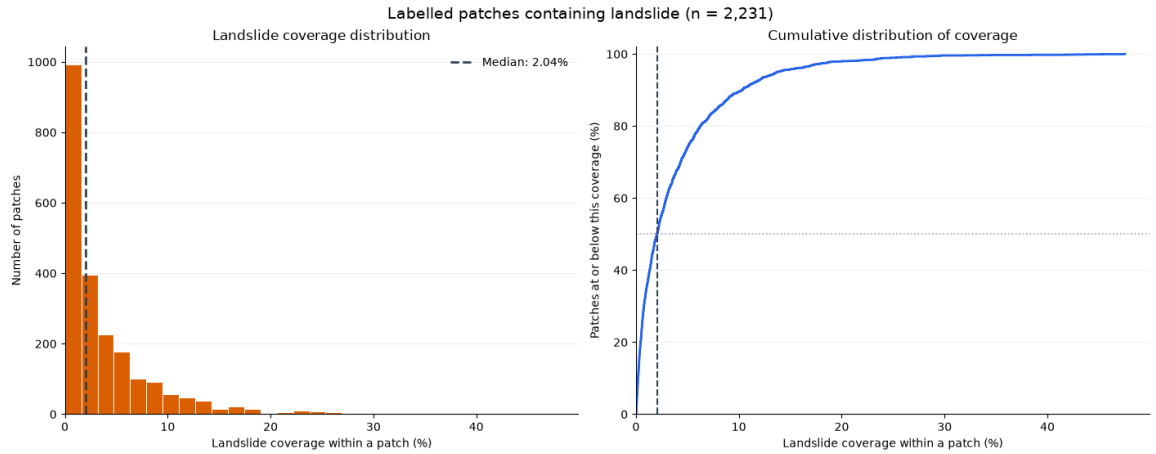


Figure 3. *Landslide coverage within the 2,231 patches that contain landslides. Left: histogram (median 2.04%, dashed line). Right: cumulative distribution. Most landslides cover only a few percent of a patch, which makes them hard to detect.*

2.3 Exploratory data analysis

Before modelling, the 14 input channels were compared between landslide and non-landslide pixels on a random development sample (25,600 pixels from 100 patches). Landslide pixels are much brighter in the visible and red-edge bands: the median of the red band B4 is 1.47 units higher for landslide pixels, followed by the short-wave infrared band B12 (+1.03), the red-edge band B5 (+0.75) and green B3 (+0.65). The near-infrared bands B6–B8 are slightly darker on landslides (−0.09 to −0.24), which matches the loss of vegetation. Landslide pixels also lie on slightly steeper (+0.25) and higher (+0.14) terrain (Figure 4).

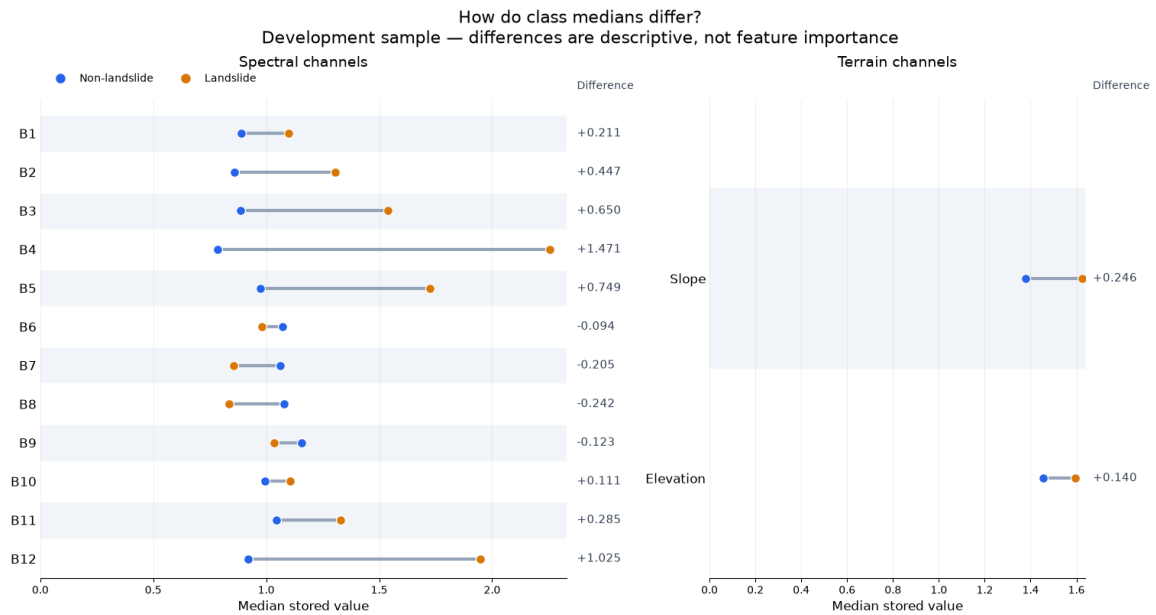


Figure 4. *Median value of each input channel for non-landslide (blue) and landslide (orange) pixels. Numbers give the difference landslide − non-landslide. Differences are descriptive and do not measure feature importance.*

Many channels carry overlapping information (Figure 5). The visible bands B2–B5 are strongly correlated with each other (Spearman ρ = 0.73–0.96), as are the near-infrared bands B6–B8 (ρ = 0.96–0.99) and the short-wave infrared bands B11 and B12 (ρ = 0.95). Elevation is strongly negatively correlated with the coastal-aerosol band B1 (ρ = −0.80). This redundancy matters for interpretation: correlated channels can replace each other in a model, so importance values must be read as group effects rather than as the effect of a single band.

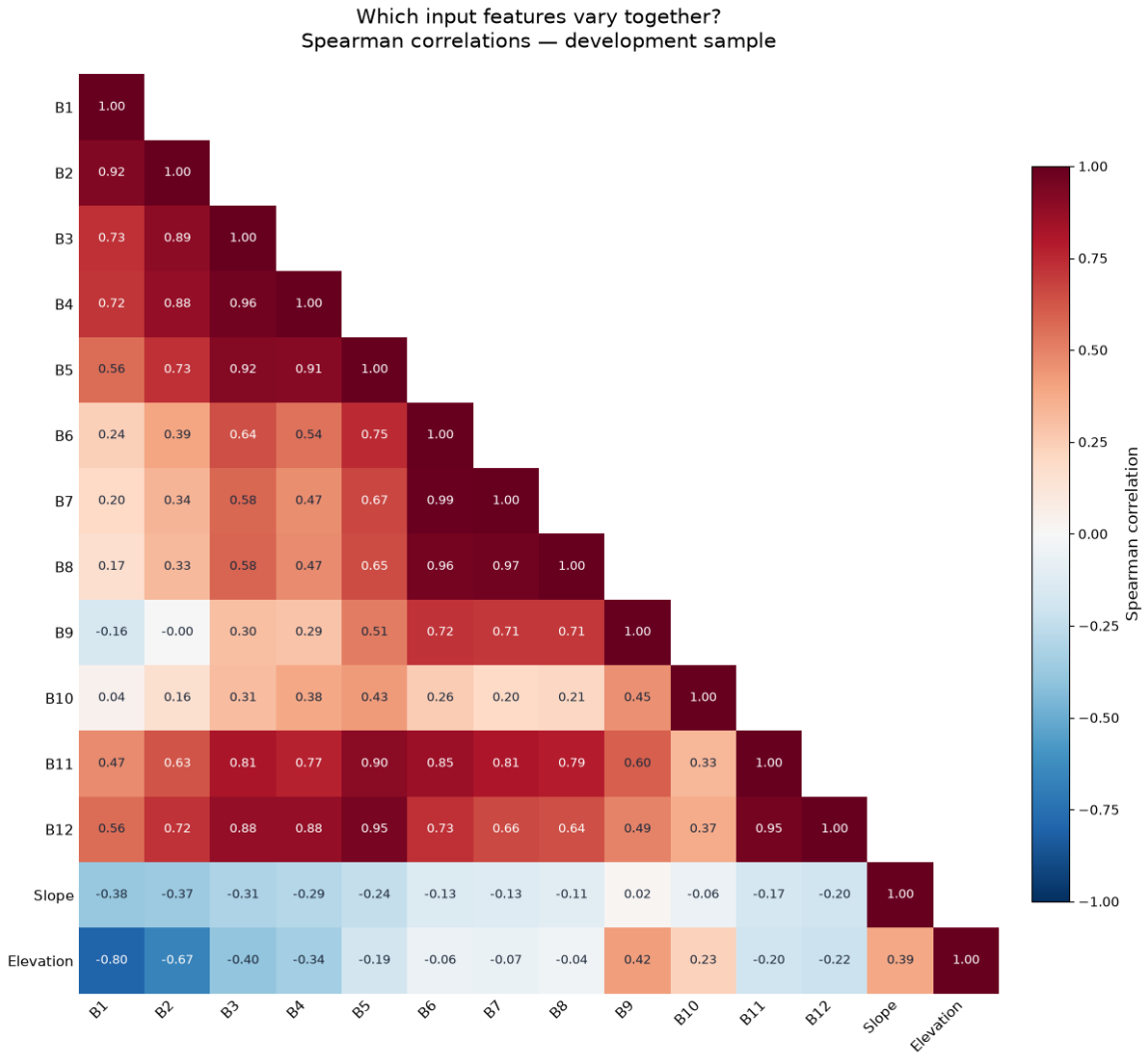


Figure 5. *Spearman correlations between the 14 input channels on the development sample. Red = positive, blue = negative correlation.*

3. Methods

Every classifier was trained, tuned and evaluated with the same protocol, the same image splits and the same metric, so that differences in results reflect the classifiers rather than the experimental setup.

3.1 Evaluation metric

The primary metric is the F2 score, which weights recall four times as heavily as precision. Missing a real landslide (a false negative) is more costly in hazard mapping than a false alarm, which can be checked by an analyst. Accuracy was not used because of the strong class imbalance.

$$F_2 = \frac{5TP}{(5TP + 4FN + FP)} = \frac{5 \times Precision \times Recall}{(4 \times Precision + Recall)}$$

Secondary metrics were precision, recall, specificity, intersection over union (IoU), ROC-AUC, average precision (AP), the Brier score, log loss and calibration error. Unless stated otherwise, metrics are pooled: confusion counts are summed over all pixels of all evaluated images before the metric is computed.

3.2 Shared evaluation protocol

1. **Image-level splitting.** All pixels of an image always stay together in training, validation or evaluation, because neighbouring pixels are strongly correlated.
2. **Nested cross-validation.** The development set was divided into 5 outer folds (607–608 images each), and each outer training part into 3 inner folds, stratified by landslide presence. All classifiers used identical fold assignments.
3. **Pixel sampling for training, full images for evaluation.** Models were trained on 512 or 1,024 uniformly sampled pixels per image (selected per classifier), which preserves the true class proportions. Every validation and evaluation image was predicted on all 16,384 pixels.
4. **Selection on inner folds only.** Hyperparameters, an optional probability calibration and the decision threshold (a grid of 101 values from 0.001 to 0.99) were chosen by the highest inner-validation F2. No specificity constraint was applied.
5. **Outer evaluation.** For each outer fold, the inner-selected configuration was refitted on the outer training images and evaluated on the held-out outer images. Each development image was therefore evaluated exactly once, by a model that never saw it during training or selection.
6. **Final model.** The same search was repeated on three development folds (2,026 training and 1,013 validation images each). The selected configuration and threshold were fitted once on all 3,039 development images and locked.
7. **Secondary test.** The locked model predicted the 760 test images once, without any further tuning. The test is called secondary because these images had already been inspected during the logistic regression analysis.

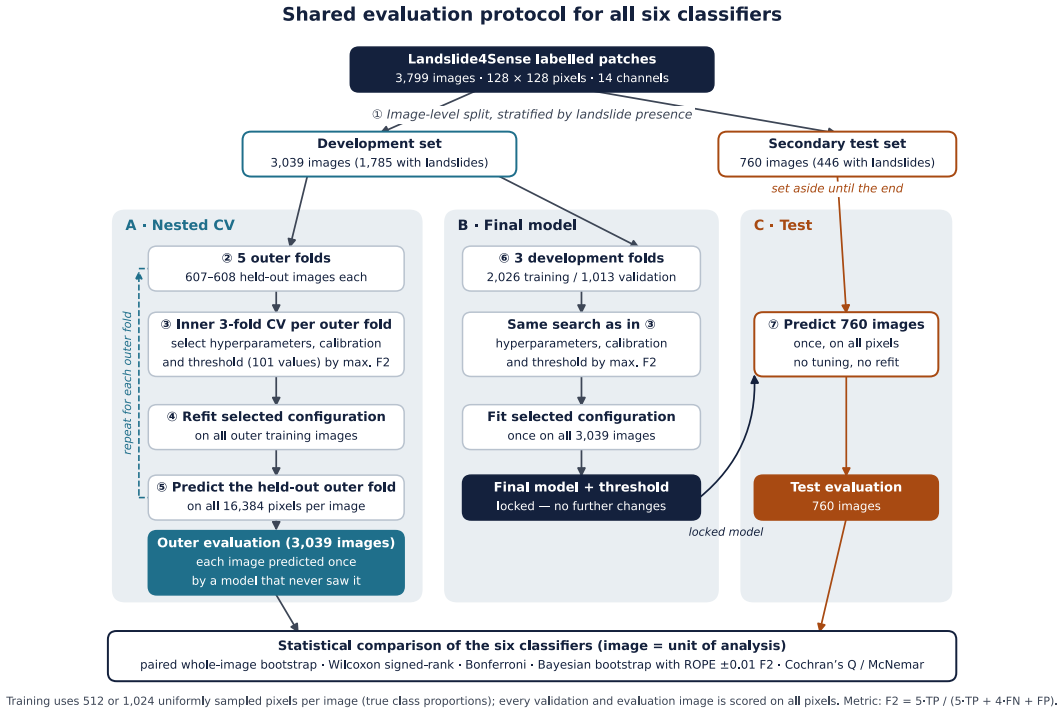


Figure 6. *Shared evaluation protocol. Circled numbers refer to the steps listed in Section 3.2.*

Uncertainty was quantified with a whole-image bootstrap: complete images were resampled with replacement (2,000–5,000 repetitions) with the fitted models held fixed. Figure 6 describes the evaluation protocols.

3.3 Classifiers

Table 2. *Selected configuration of each classifier.*

Classifier	Selected configuration	Training pixels per image	Calibration	Threshold
Logistic regression (LR)	Standardised inputs, L2 penalty, C = 0.1, no class weighting	512	Sigmoid (Platt)	0.10
LR + interactions	14 inputs + 91 pairwise products (105 predictors), standardised, C = 1	512	None	0.11
Decision tree	Max. depth 15, min. 5 samples per leaf, entropy, balanced class weights, cost-complexity pruning 10^{-5} (3,032 leaves)	1,024	None	0.92
Random forest	200 trees, max. depth 20, min. 10 samples per split, 50% of features per split, entropy, 50% bootstrap samples	1,024	None	0.17
HGB	150 iterations, learning rate 0.1, 63 leaves per tree, min. 100 samples per leaf, L2 = 1	1,024	None	0.13
XGBoost	600 trees, depth 4, learning rate 0.05, min. child weight 10, 80% row and column subsampling, L2 = 10	1,024	None	0.14

The search effort differed by classifier: 25 random configurations plus a local simplification check for the decision tree, 8 configurations plus tree-count, sample-size and calibration checks for the random forest, 6 configurations for HGB and 3 for XGBoost. Sigmoid and isotonic calibration were compared for logistic regression, the decision tree and the random forest; only raw-feature LR selected one.

3.4 Diagnostics and explainability

Each locked model was analyzed with ROC and precision–recall curves, reliability diagrams (15 uniform bins, with expected and maximum calibration error, ECE as expected calibration error and MCE as maximum calibration error), per-image metrics, results by landslide coverage, and maps of six fixed test images. Feature reliance was measured by permutation importance: the drop in F2 when one input channel is shuffled. For XGBoost, exact TreeSHAP values were computed to show how each channel pushes individual pixel predictions towards or away from "landslide".

3.5 Statistical comparison

The comparison treats each **image** as one observation and uses a **paired** design, because all classifiers were evaluated on the same images. The null hypothesis of each test was that two classifiers perform equally (two-sided, $\alpha = 0.05$); all decisions were fixed before the results were seen.

- **Paired whole-image bootstrap** (5,000 resamples, identical for all models) of the pooled F2 difference, giving a 95% confidence interval, a p-value and the share of resamples in which one model is better.

- **Wilcoxon signed-rank test** of per-image F2 differences on images that contain landslides. A paired t-test was ruled out because Shapiro–Wilk tests rejected normality of the differences in every comparison with the reference model ($p < 10^{-15}$).
- **Bonferroni correction** for the 15 pairwise comparisons per evaluation set ($\alpha = 0.0033$).
- **Bayesian bootstrap** (Dirichlet-weighted images, 5,000 posterior draws) with a **region of practical equivalence (ROPE) of ±0.01 F2**, smaller than the fold-to-fold variation of each model (SD ≈ 0.015). It estimates the probability that two classifiers are practically equivalent — a question a non-significant p-value cannot answer.
- **Image-level tests** of two yes/no outcomes (landslide image detected; landslide-free image without any false alarm): Cochran's Q across all six classifiers, exact McNemar tests against the reference model, and Beta(1, 1) posteriors.

The reference model, the random forest, was chosen as the best model on the outer evaluation, without using the test set. Pixel-level tests such as chi-square on pixel counts were not used: with 12 million dependent pixels they would declare negligible differences significant.

All analyses were run in Python 3.13 with scikit-learn 1.9.0, XGBoost 3.4.1, NumPy 2.4.6 and pandas 3.0.3, with random seed 43.

4. Results

4.1 Results of the individual classifiers

This section reports the selection and the results of each classifier separately, in the order in which they were developed. Sections 4.2–4.7 then compare all six classifiers directly on identical images.

4.1.1 Logistic regression on the raw features

A first diagnostic on the training part of outer fold 1 tested whether the terrain layers add information to the spectral bands. Terrain alone was nearly useless (inner-validation F2 0.143), spectral bands alone reached 0.584, and both together reached 0.627 (Figure 7). All 14 channels were therefore kept for every classifier.

The selected model uses standardised inputs, C = 0.1, no class weighting, 512 training pixels per image, sigmoid calibration and a threshold of 0.10. Its outer F2 was 0.629 [0.614, 0.643], with all five folds between 0.61 and 0.65. The coefficients of the final model (Figure 8) show that it relies on contrasts between neighbouring bands rather than on single bands: the largest positive weights are B6 (+2.31), B3 (+2.03) and B12 (+1.25), the largest negative weights are B7 (−4.14), B2 (−1.98) and B11 (−1.45). Strongly correlated pairs such as B6 and B7 ($\rho = 0.99$) receive opposite signs, so the coefficients describe conditional effects and should not be read as independent importance. The red band B4, the clearest single difference between the classes in Figure 4, receives only +0.14 because its information is shared with the other visible bands.

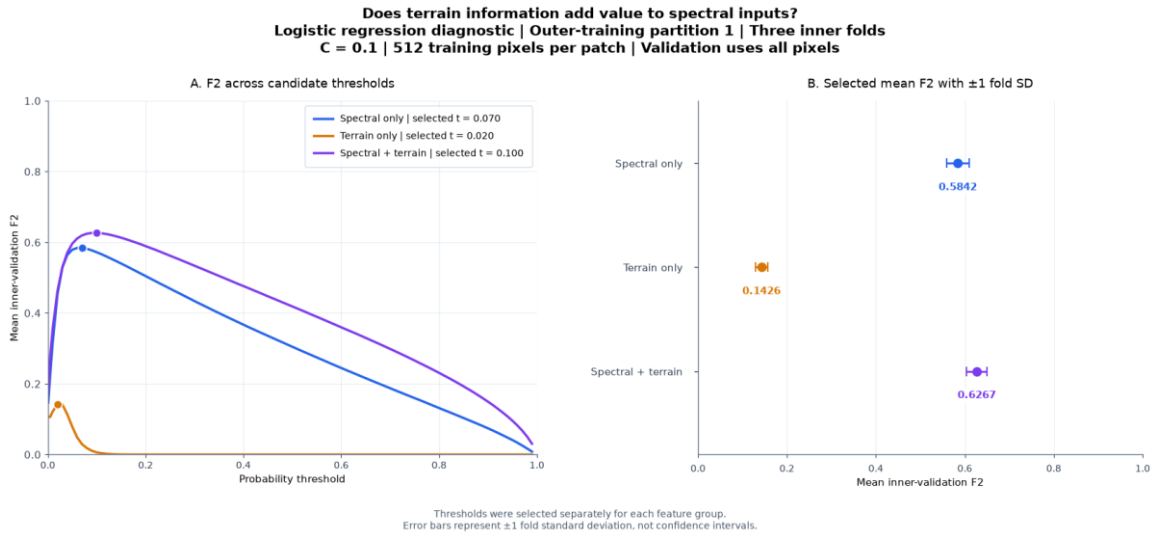


Figure 7. *Logistic regression with spectral bands only, terrain only, and both. A: inner-validation F2 across thresholds. B: selected F2 ± 1 fold SD. Terrain adds about +0.04 F2 to the spectral bands.*

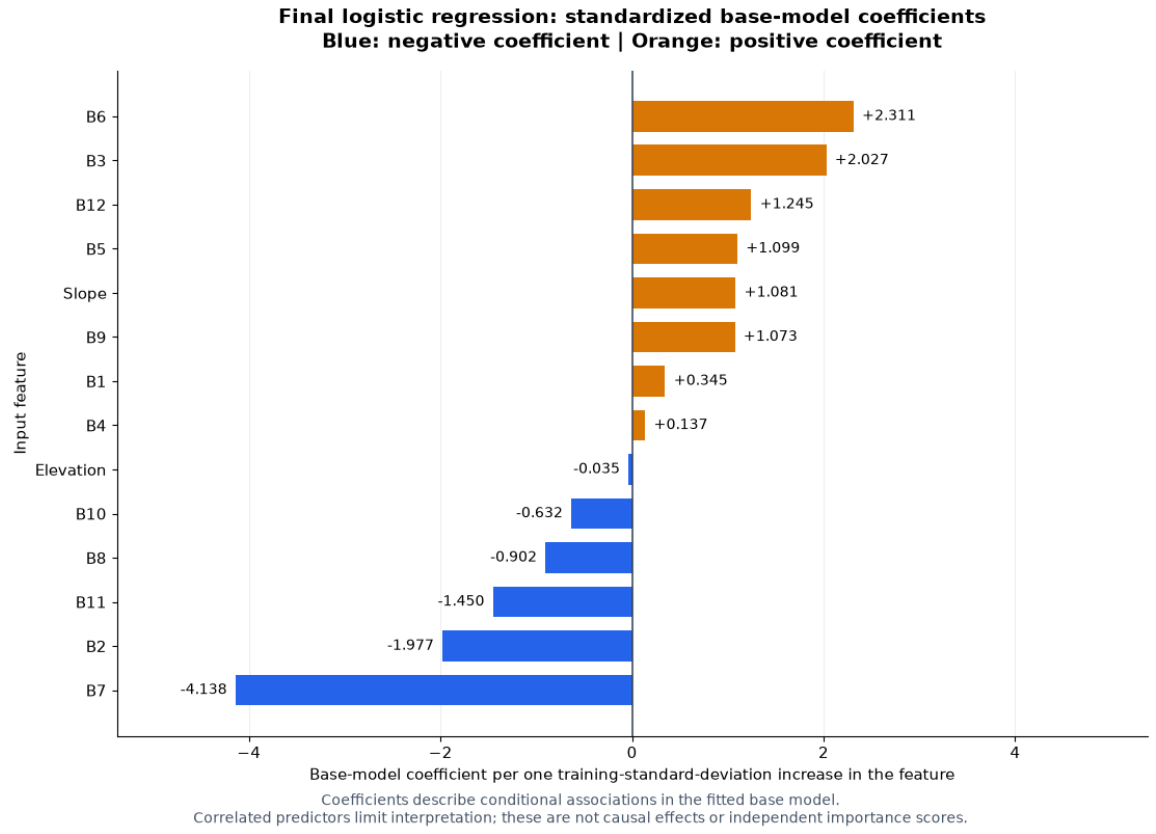


Figure 8. *Standardised coefficients of the final logistic regression. Orange = positive (towards landslide), blue = negative.*

On the secondary test set the model found 74.1% of the landslide pixels (196,538 of 265,381) but also flagged 348,771 non-landslide pixels, 2.9% of all background pixels. With a precision of 0.36, almost two of three predicted landslide pixels were false alarms, giving a test F2 of 0.612 (Figure 9). A single linear boundary in the 14-dimensional feature space is clearly too simple for this problem.

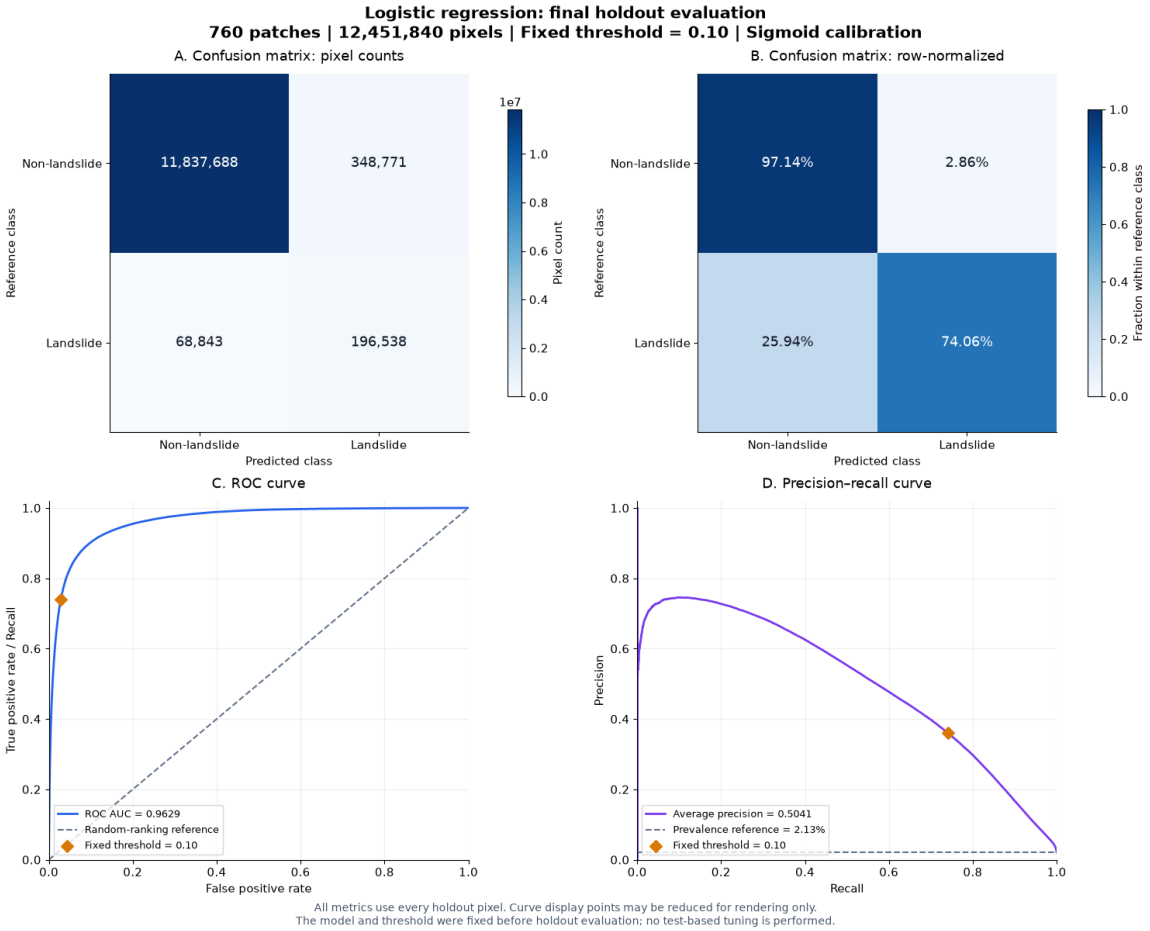


Figure 9. *Final logistic regression on the 760 test images. A–B: confusion matrix as pixel counts and as row percentages. C–D: ROC and precision–recall curves with the locked operating point.*

4.1.2 Logistic regression with pairwise interactions

To test whether the weakness of LR comes from its linear form, the 91 pairwise products of the 14 standardised inputs were added (105 predictors). This extension was designed after the first LR results had been inspected and is therefore exploratory. The selected model uses C = 1, no calibration and a threshold of 0.11.

The gain was large and stable: in every outer fold the interaction model was better by +0.117 to +0.122 F2 (Figure 10), raising the outer F2 from 0.629 to 0.748.

On the test set, recall rose from 74.1% to 83.8% (222,418 landslide pixels found) while false-positive pixels fell by 39%, from 348,771 to 212,796. The test F2 rose from 0.612 to 0.743. The improvement was largest for the smallest landslides: in images with at most 1% landslide coverage, F2 rose from 0.313 to 0.473, compared with 0.839 to 0.890 in images with more than 15% coverage. False-alarm areas in landslide-free images shrank as well (Figure 11). The spatial examples in Figure 12 show cleaner, more compact detections, but bright riverbeds remain a source of false alarms.

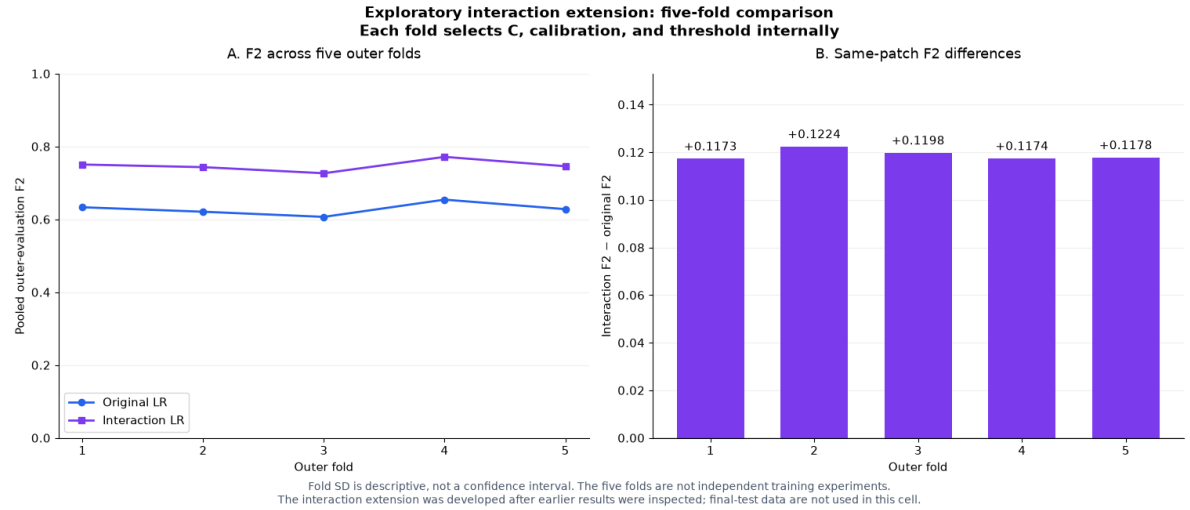


Figure 10. *Original and interaction logistic regression on the five outer folds. A: F2 per fold. B: same-image F2 difference per fold.*

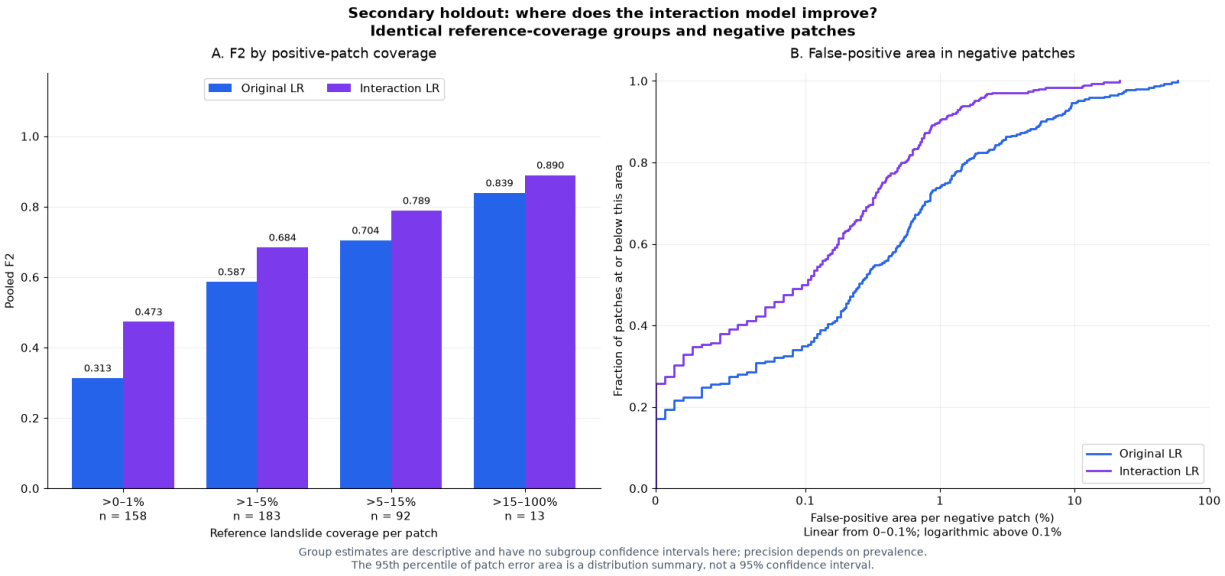


Figure 11. *Original versus interaction LR on the test set. A: F2 by landslide coverage group. B: cumulative distribution of the false-positive area in landslide-free images (further left = fewer false alarms).*

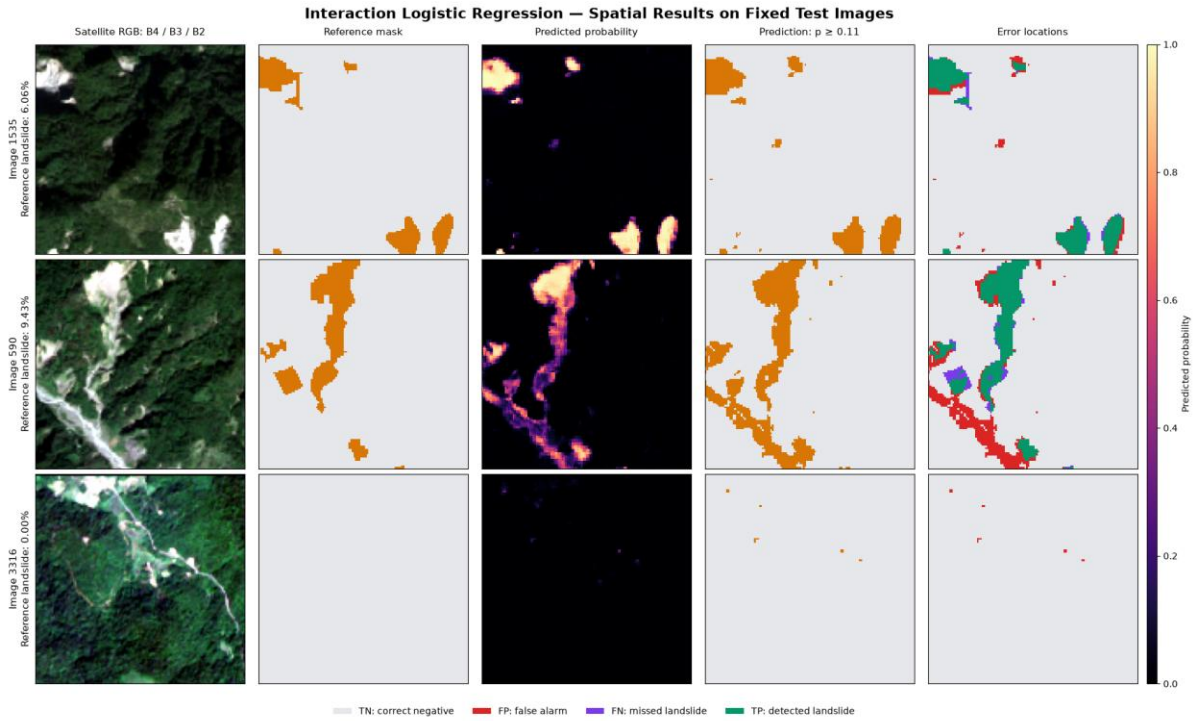


Figure 12. *Interaction logistic regression on three fixed test images. Columns: true-colour image, reference mask, predicted probability, prediction at threshold 0.11, error map (green = detected, red = false alarm, blue = missed).*

4.1.3 Decision tree

Twenty-five random configurations were searched, followed by a check of simpler trees. The final pipeline was chosen among two sampling sizes and three calibration options on three development splits (Figure 13). The uncalibrated tree trained on 1,024 pixels per image won with a mean validation F2 of 0.734 at a threshold of 0.92, narrowly ahead of sigmoid (0.730) and isotonic calibration (0.730). The final tree has a maximum depth of 15, uses entropy, balanced class weights and light pruning, and ends in 3,032 leaves. Figure 14 shows its first three levels.

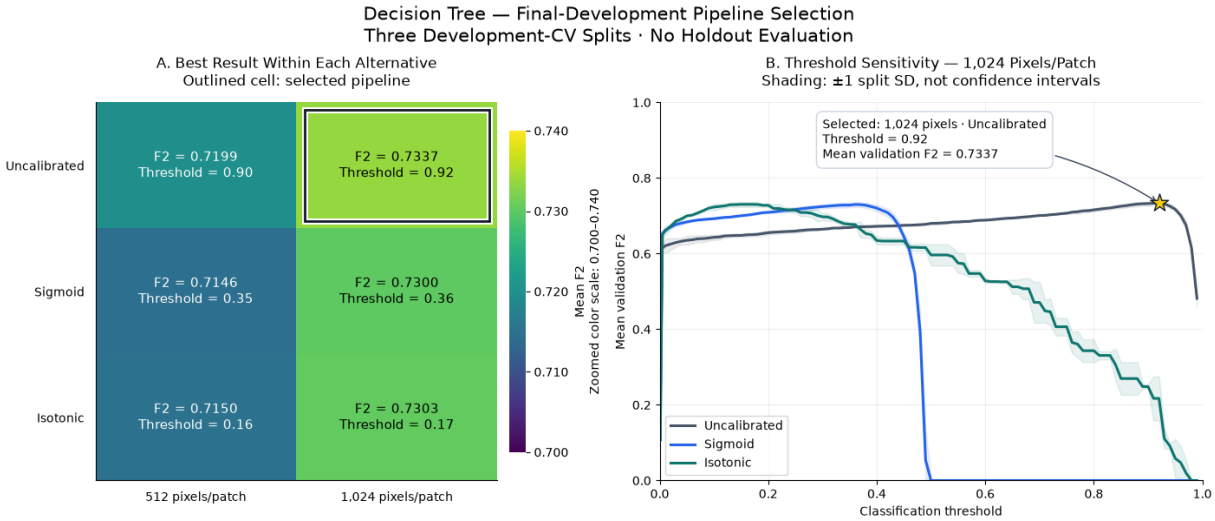


Figure 13. *Final-development selection of the decision tree. A: best F2 for each combination of sampling size and calibration (outlined = selected). B: validation F2 across thresholds for the 1,024-pixel pipelines; shading = ± 1 split SD.*

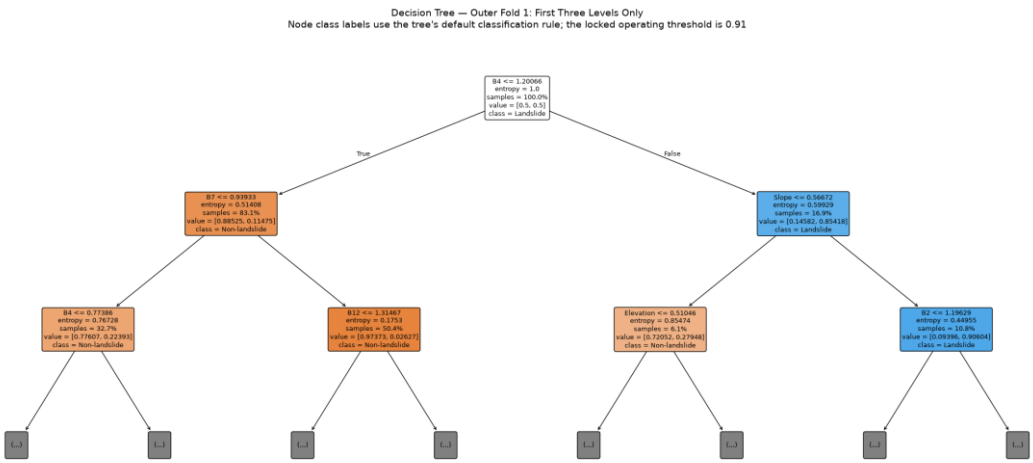


Figure 14. *The first three levels of the decision tree of outer fold 1 (orange = majority non-landslide, blue = majority landslide).*

The unusual threshold of 0.92 follows from the balanced class weights, which push the raw scores of landslide-like leaves towards 1. The scores therefore strongly overestimate the true landslide probability (Figure 17), and only very confident leaves are worth flagging. This also makes the decision tree the worst-calibrated model (Brier score 0.036, ECE 0.043, compared with ≤ 0.014 and ≤ 0.004 for all others). The tree overfits visibly: in the inner folds of outer fold 1 its training F2 was 0.82–0.83, its validation F2 only 0.70–0.74.

The tree depends heavily on one band. B4 accounts for 57% of the impurity reduction, and permuting it lowers F2 by 0.53. Elevation (0.18), B7 (0.16) and slope (0.13) follow far behind; both importance measures agree on the four leading inputs (Figure 15).

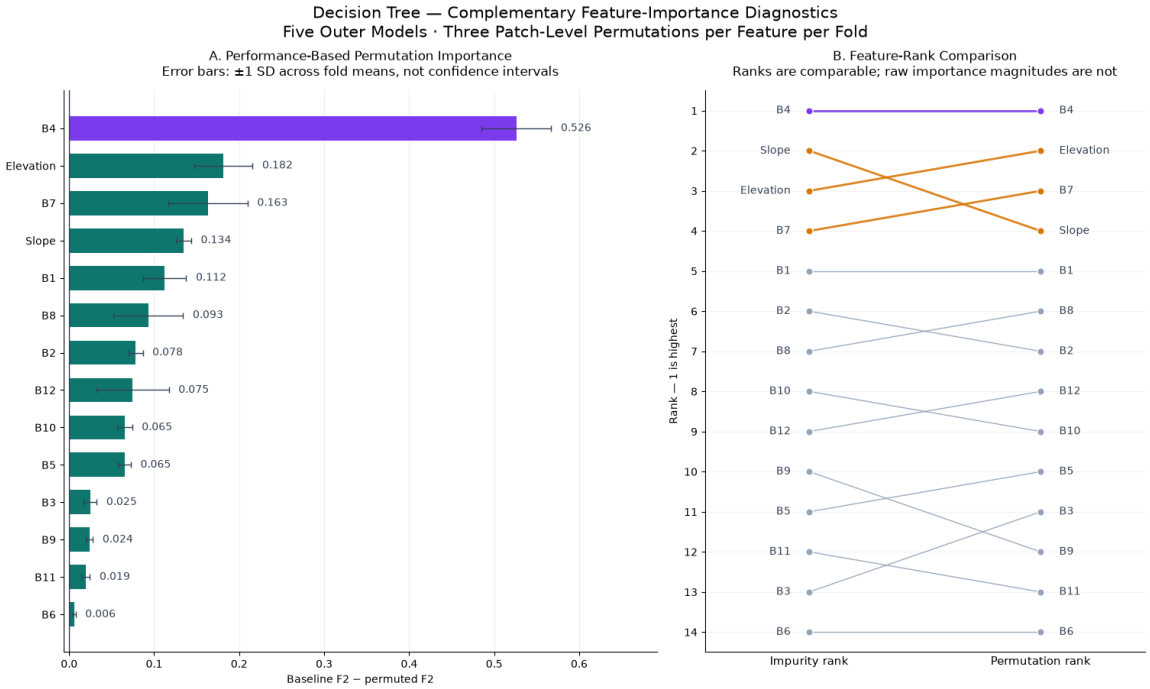


Figure 15. *Decision tree feature importance across the five outer models. A: permutation importance (drop in F2; bars = ± 1 SD across folds). B: rank agreement between impurity-based and permutation importance.*

The outer F2 was 0.738. On the test set the tree found 79.6% of the landslide pixels (211,170) and produced 188,101 false-positive pixels, giving F2 = 0.723 (Figure 16). Compared with interaction LR, the tree raises fewer false alarms but misses more landslide pixels.

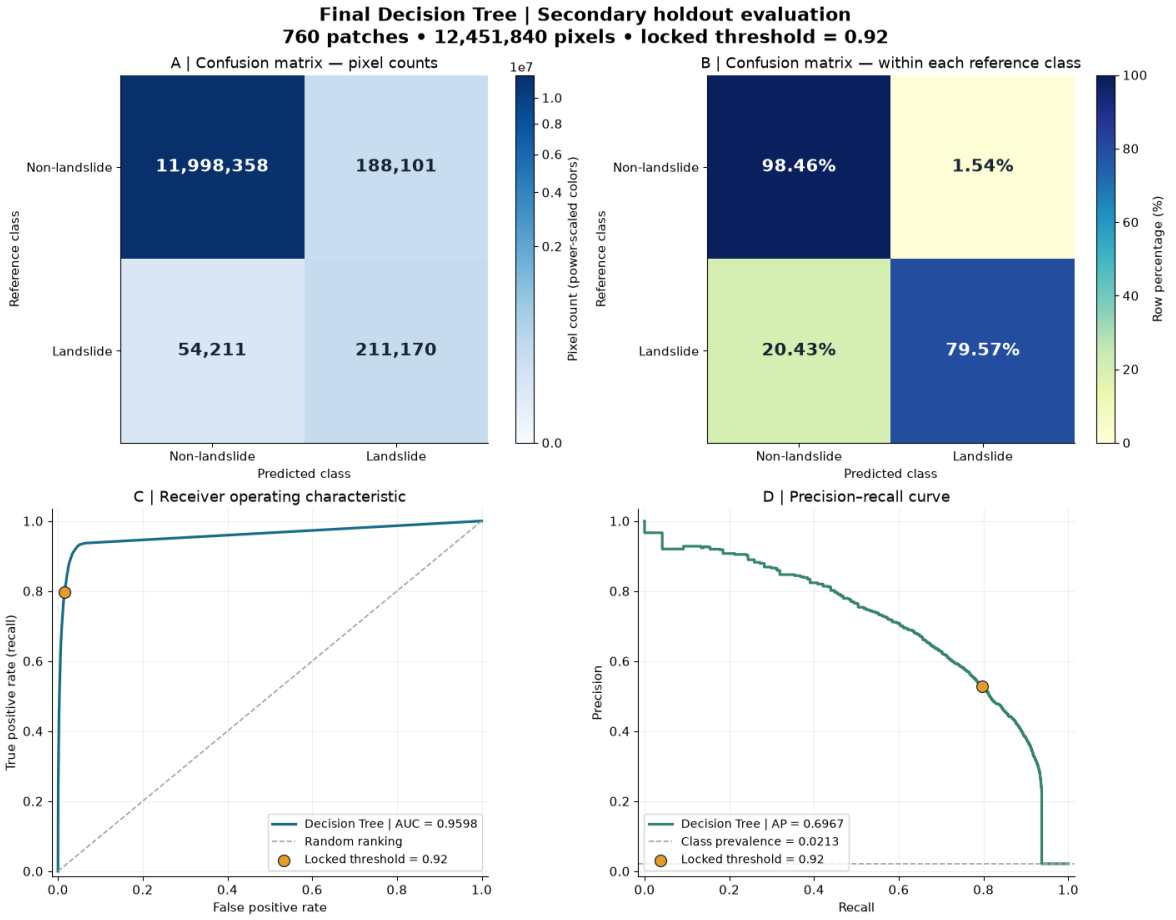


Figure 16. *Final decision tree on the 760 test images: confusion matrix (counts and row percentages), ROC curve and precision–recall curve with the locked threshold 0.92.*

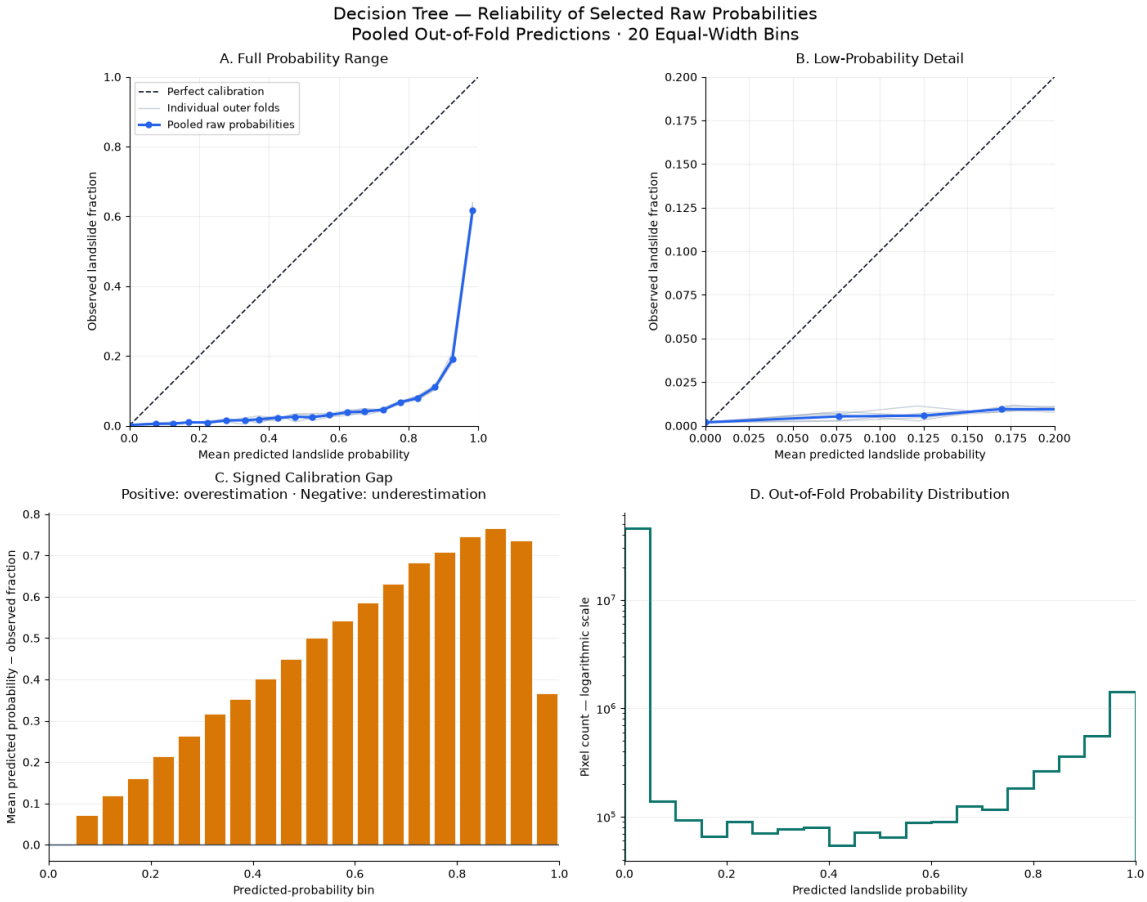


Figure 17. *Reliability of the raw decision-tree scores (pooled out-of-fold predictions, 20 bins). The curve lies far below the diagonal: the scores overestimate the landslide probability.*

4.1.4 Random forest

Eight configurations were searched, followed by checks of the number of trees, the training sample size and calibration. More trees helped only slightly: going from 50 to 100 trees raised the inner-validation F2 by 0.22 percentage points, and from 50 to 200 trees by 0.32 points (Figure 18). The final forest has 200 trees of maximum depth 20, considers 50% of the features at each split, draws 50% bootstrap samples and uses a threshold of 0.17 without calibration.

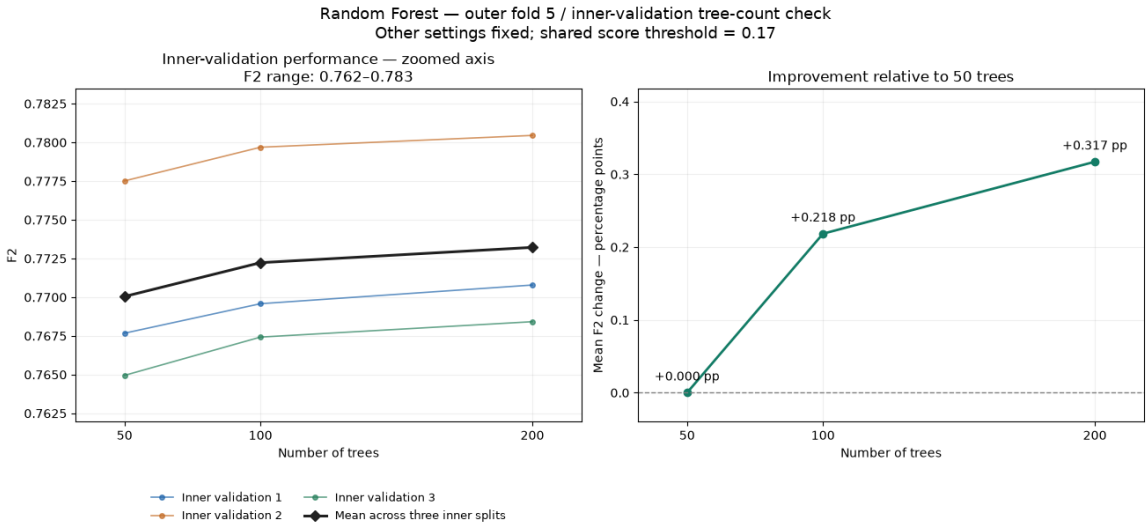


Figure 18. *Effect of the number of trees on inner-validation F2 (outer fold 5, other settings fixed). Gains flatten after 100 trees.*

The random forest reached the highest pooled outer F2 of all classifiers, 0.776 [0.768, 0.784], with very similar values in all five folds (Figure 19). On the 1,254 development images without landslides, 363 images (29%) received no false-positive pixel at all, and another 706 images (56%) received at most 100 false-positive pixels, i.e. at most 0.6% of the image. Only 6 images (0.5%) had more than 1,000 false-positive pixels (Figure 20). Most false alarms are therefore small specks rather than large wrongly mapped areas.

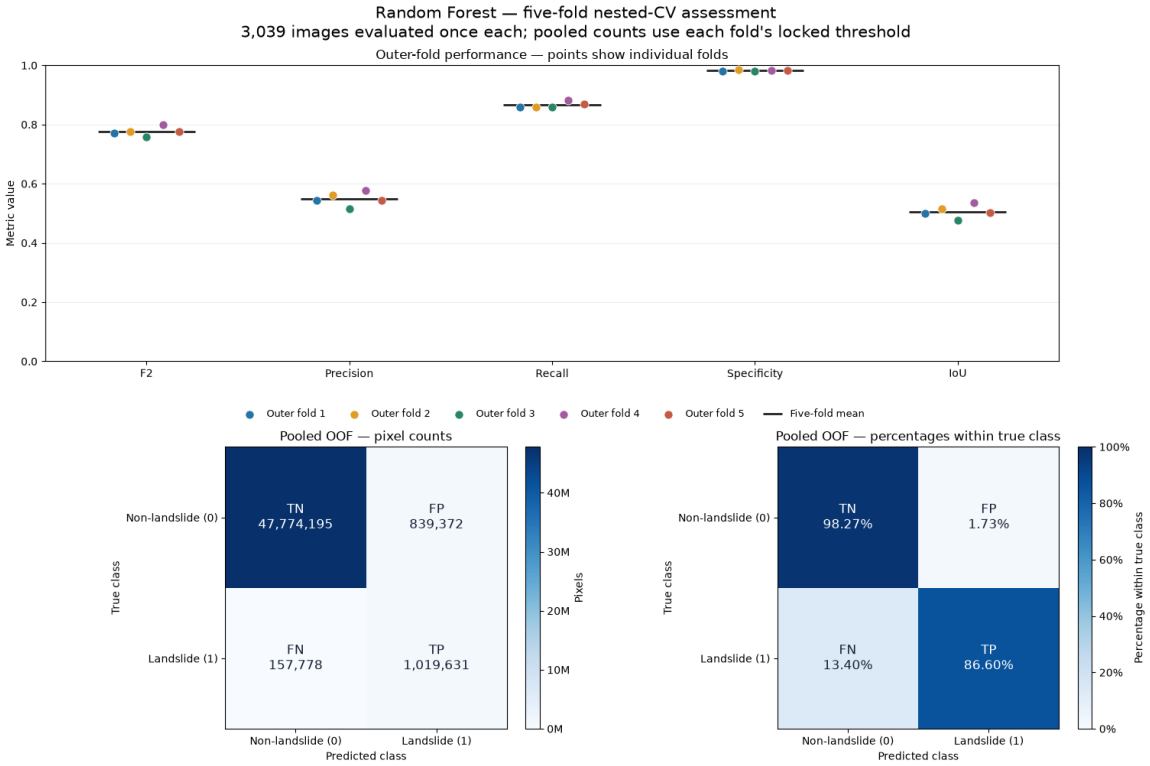


Figure 19. *Random forest five-fold nested-CV results. Top: metrics per fold (coloured points) and their mean. Bottom: pooled out-of-fold confusion matrix as counts and row percentages.*

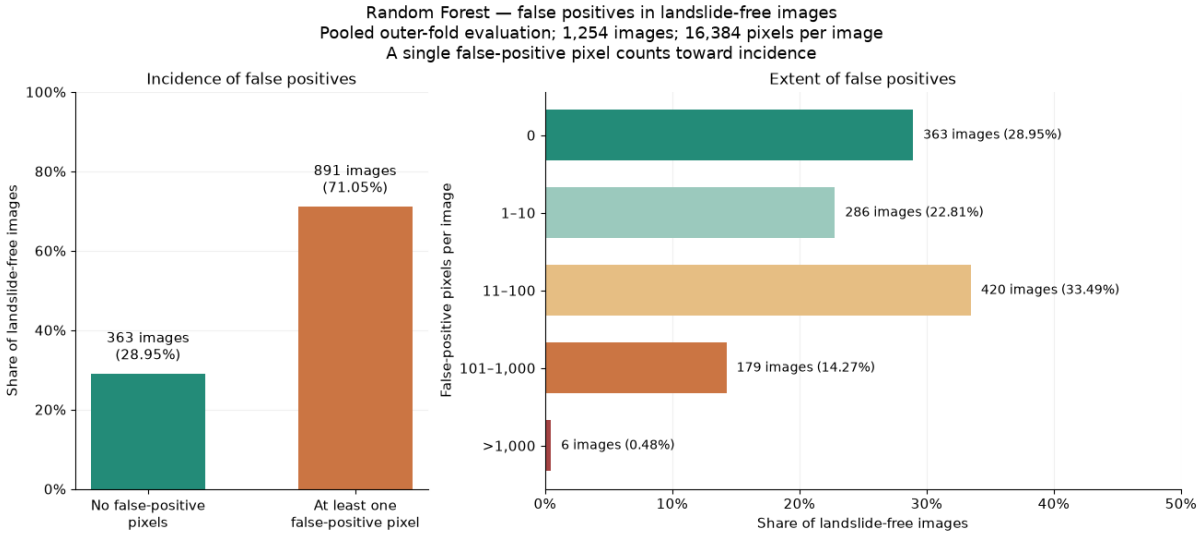


Figure 20. *False positives in the 1,254 landslide-free development images. Left: share of images with and without any false-positive pixel. Right: number of false-positive pixels per image.*

Permutation importance (Figure 21) ranks B4 first (F2 drop 24.0 percentage points), followed by elevation (19.6) and slope (13.5), then B12 (9.6), B1 (8.5) and B7 (8.4). Impurity-based importance also ranks B4 first (24.7% of the impurity reduction) but places the correlated visible bands B3 and B2 higher than permutation does, a known bias of impurity importance towards correlated inputs.

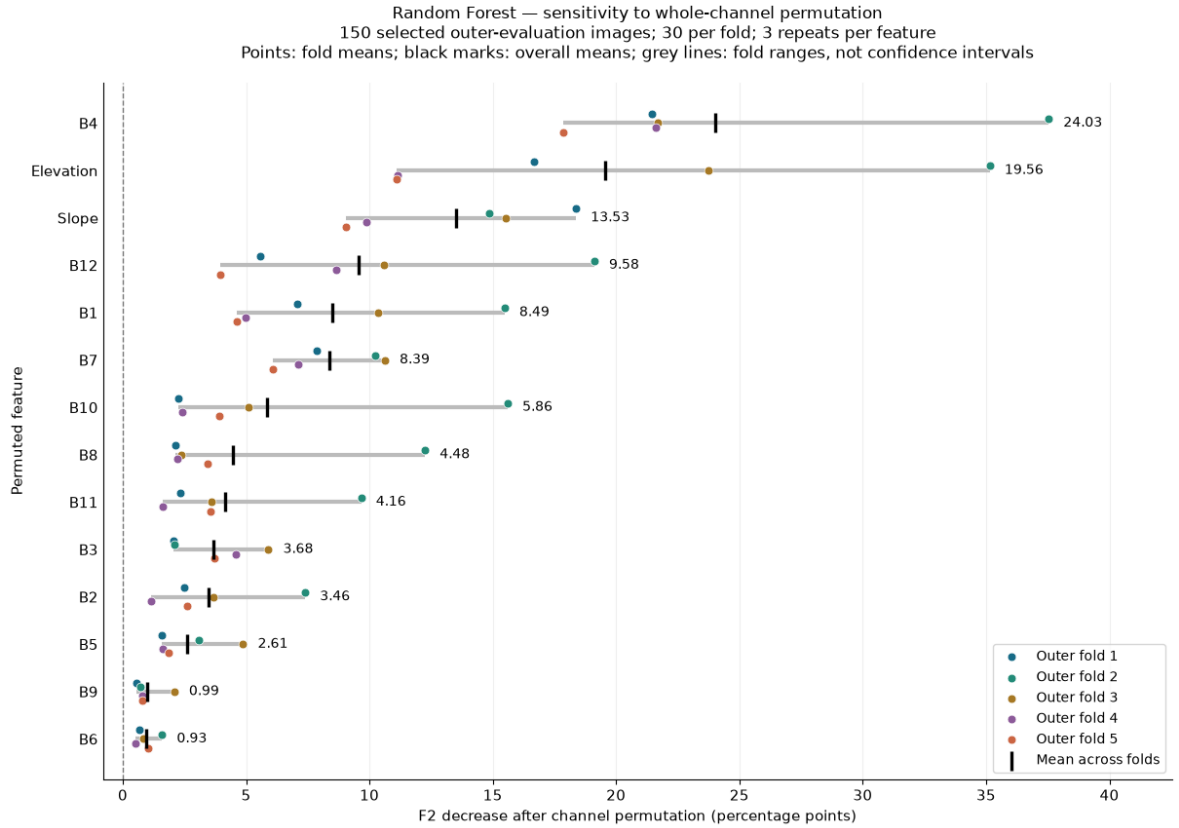


Figure 21. *Random forest permutation importance across the five outer models (F2 drop in percentage points; coloured points = folds, black marks = mean).*

On the secondary test set the random forest found 86.0% of the landslide pixels (228,223 of 265,381) and flagged 198,381 background pixels, 1.6% of all non-landslide pixels, giving F2 = 0.767 (Figure 22). The spatial examples (Figure 23) show compact, well-delineated detections; the errors lie mainly along landslide edges and on bright river channels. Performance grows steadily with landslide size: F2 rises from about 0.52 in the 158 test images with at most 1% landslide coverage to about 0.90 in the 8 images with more than 20% coverage (Figure 24).

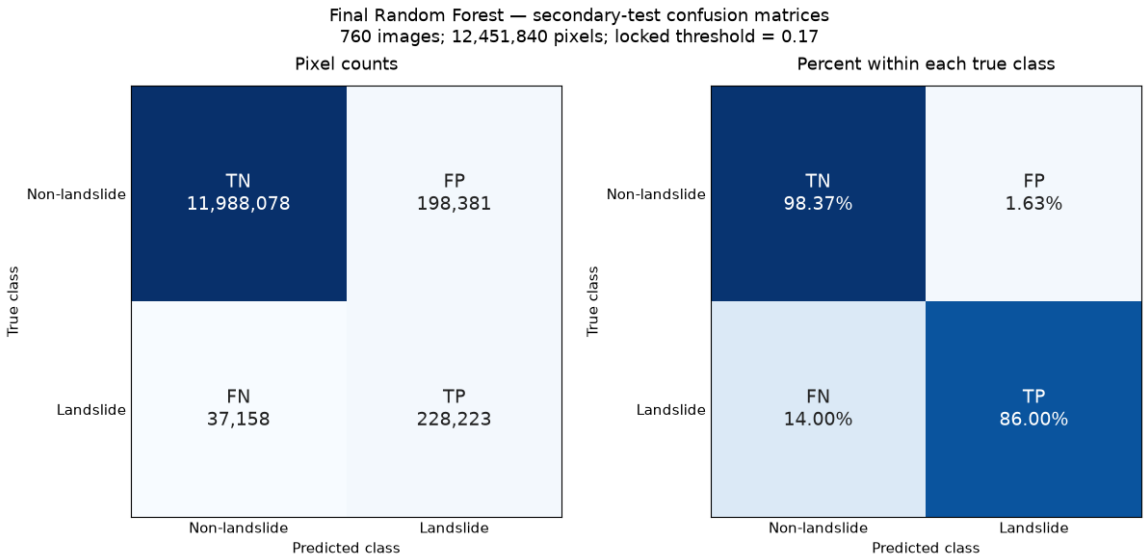


Figure 22. *Final random forest on the 760 test images: confusion matrix as pixel counts and as percentages within each true class (threshold 0.17).*

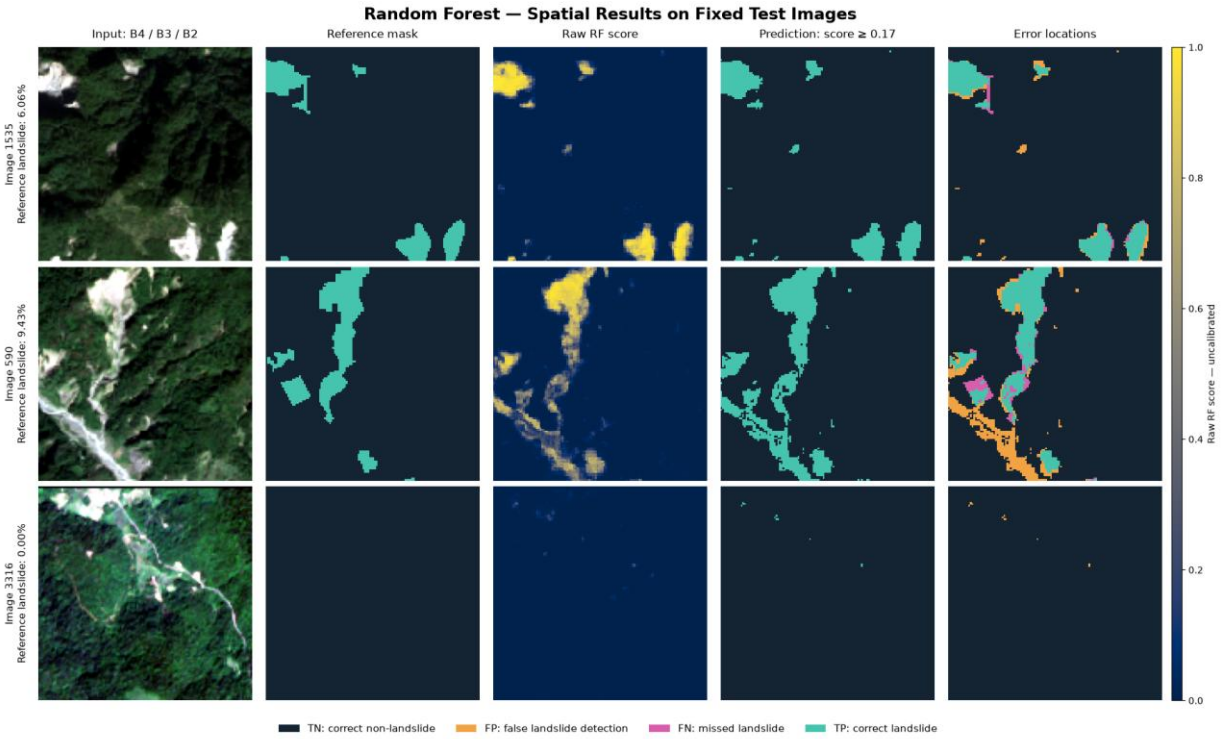


Figure 23. *Random forest on three fixed test images. Columns: true-colour image, reference mask, raw score, prediction at threshold 0.17, error locations.*

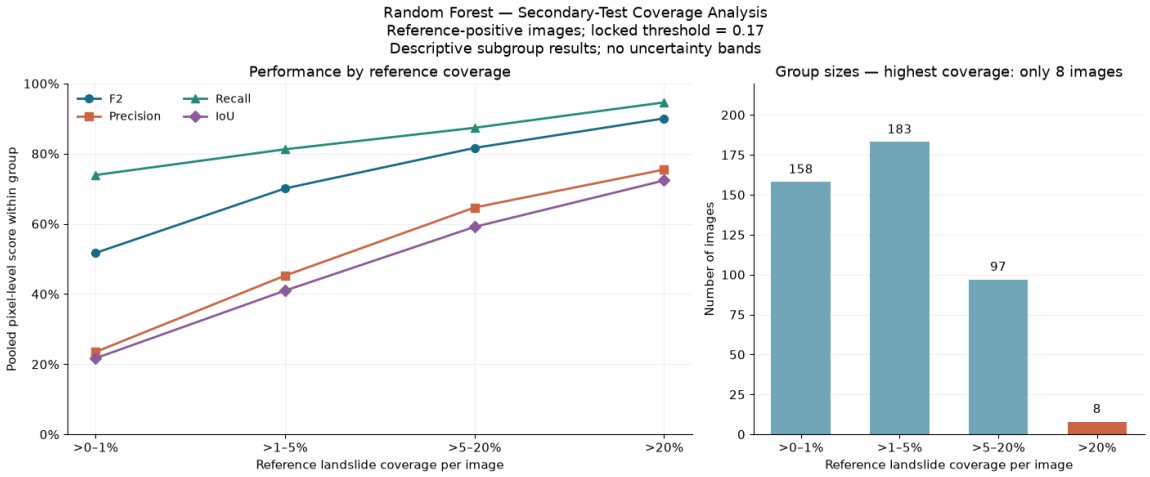


Figure 24. *Random forest test performance by landslide coverage per image. Left: F2, precision, recall and IoU per group. Right: number of images per group.*

The forest was the most expensive model to build: each refit on the outer training data took 8–11 minutes, and the full nested search took roughly 40 hours.

4.1.5 Histogram gradient boosting (HGB)

Six candidate configurations were evaluated with 114 base-model fits. All candidates were close (pooled inner-validation F2 0.764–0.774), which shows that HGB is not sensitive to its settings in this range. Candidate 6 was selected: 150 iterations, learning rate 0.1, 63 leaves per tree, at least 100 samples per leaf, L2 regularisation 1, and a threshold of 0.13 (Figure 25).

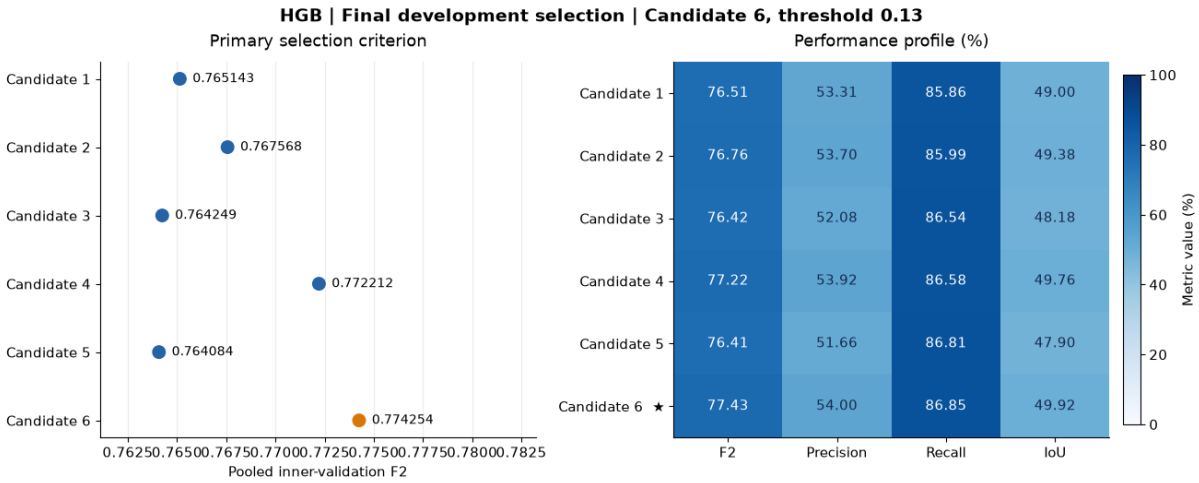


Figure 25. *HGB final development selection. Left: pooled inner-validation F2 of the six candidates (orange = selected). Right: F2, precision, recall and IoU of each candidate (%).*

The pooled outer F2 was 0.774. The folds ranged from 0.755 (fold 3) to 0.796 (fold 4), and the whole-image bootstrap distribution of the pooled F2 is narrow and symmetric (Figure 26). On the test set HGB found 86.4% of the landslide pixels (229,339), the highest recall of all classifiers, while flagging 205,505 background pixels (1.7%), for F2 = 0.766 (Figure 27).

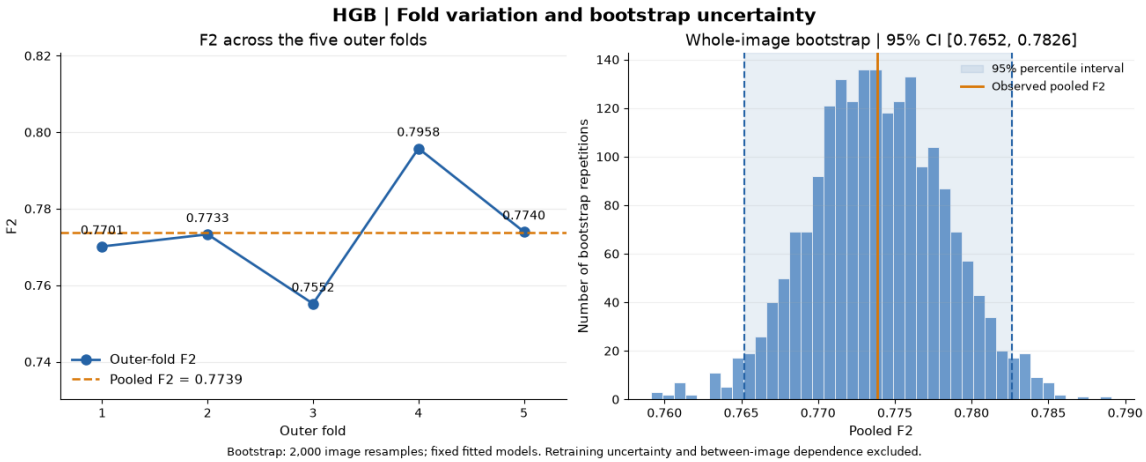


Figure 26. *HGB outer evaluation. Left: F2 per outer fold with the pooled F2 (dashed). Right: whole-image bootstrap distribution of the pooled F2 with its 95% interval.*

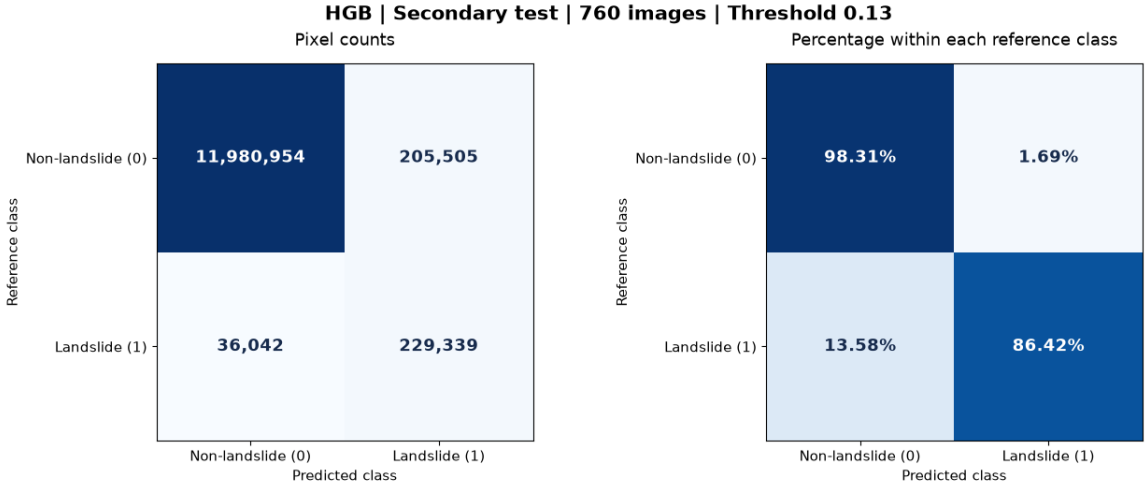


Figure 27. *Final HGB on the 760 test images: confusion matrix as pixel counts and as percentages within each reference class (threshold 0.13).*

HGB ranks pixels very well (ROC-AUC 0.991, average precision 0.774), and its probabilities are well calibrated (ECE 0.0018), although its largest bin error (MCE 0.087) is higher than that of the random forest (Figure 28). The six diagnostic test images in Figure 29 show the same pattern as for the other ensembles: compact scars are found reliably, while bright riverbeds produce most false alarms.

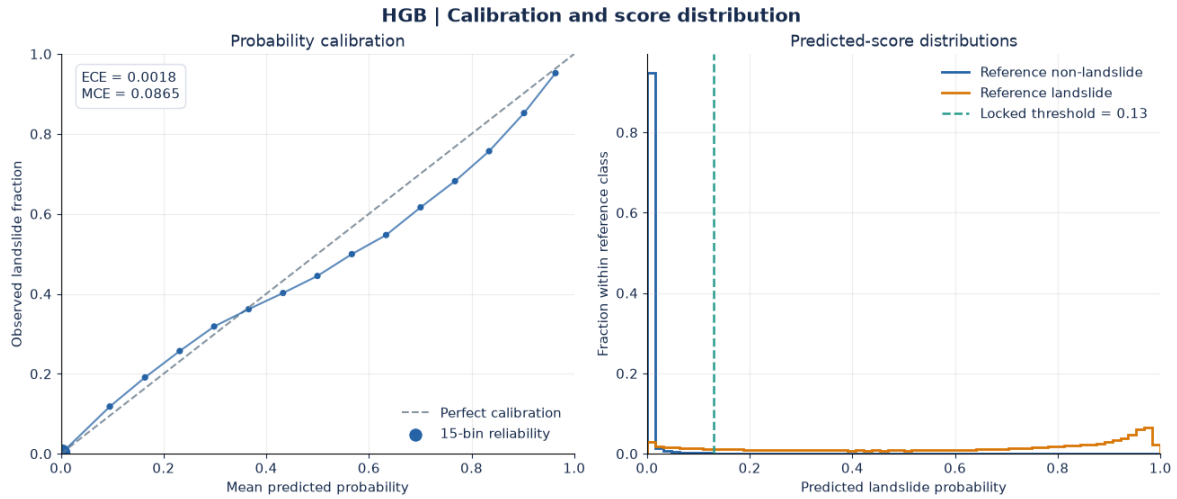


Figure 28. *HGB calibration on the test set. Left: reliability diagram (15 bins). Right: predicted-score distributions of landslide and non-landslide pixels with the locked threshold.*

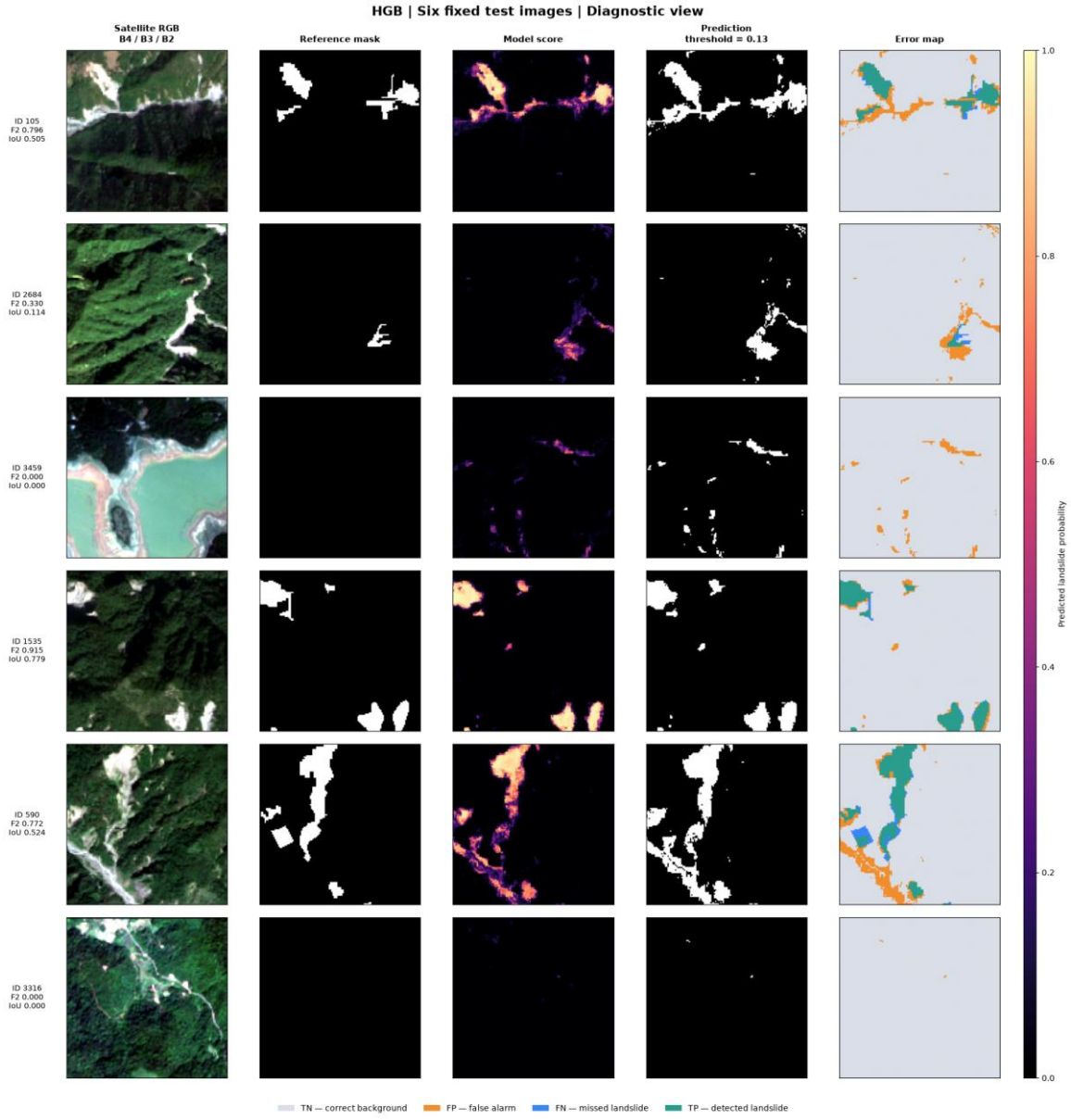


Figure 29. *HGB on six fixed test images. Columns: true-colour image, reference mask, predicted probability, prediction, error map.*

Permutation importance (Figure 30) confirms the dominance of the red band: permuting B4 lowers F2 by 0.41, followed by B1 (0.19), elevation (0.18), slope and B7 (0.11 each). The final model was fitted in 63 seconds.

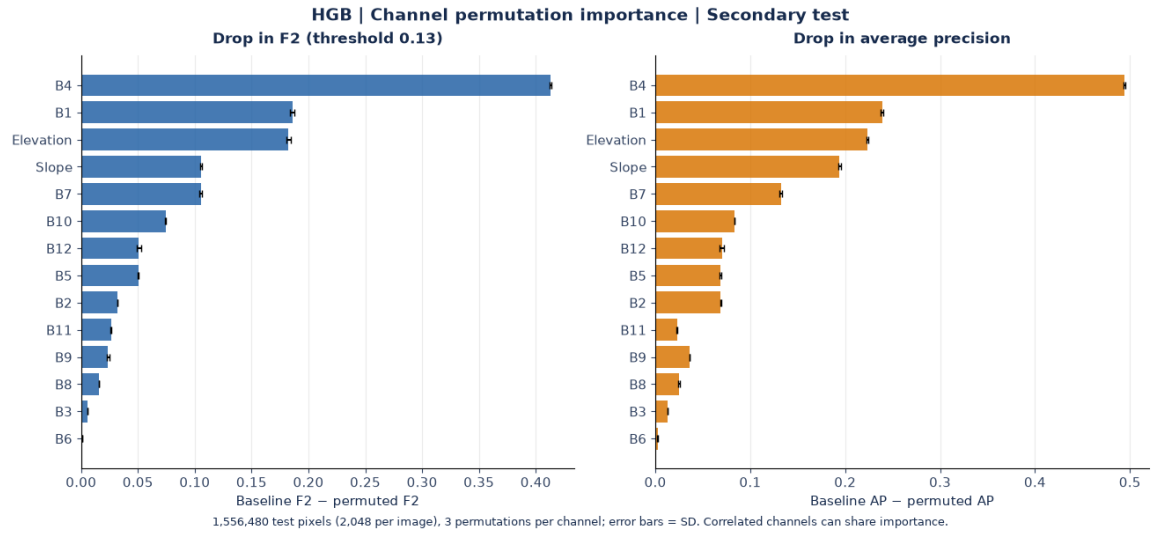


Figure 30. *HGB channel permutation importance on the test set. Left: drop in F2. Right: drop in average precision. Error bars = SD over three permutations.*

4.1.6 XGBoost

Three candidate configurations were compared. Candidate 2 (600 trees of depth 4, learning rate 0.05, 80% row and column subsampling, L2 regularisation 10) reached the highest pooled inner-validation F2, 0.773 at a threshold of 0.14 (Figure 31).

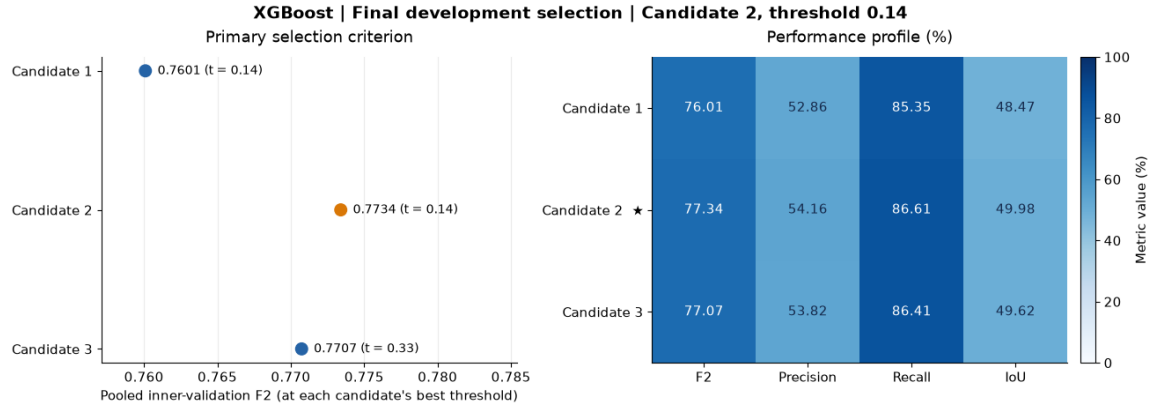


Figure 31. *XGBoost final development selection. Left: pooled inner-validation F2 of the three candidates at each candidate's best threshold. Right: performance profile (%).*

The pooled outer F2 was 0.773, with folds between 0.754 and 0.795 (Figure 32). The fold pattern is almost identical to that of HGB — fold 3 is the hardest and fold 4 the easiest for both — which shows that fold-to-fold variation reflects the images, not the algorithm.

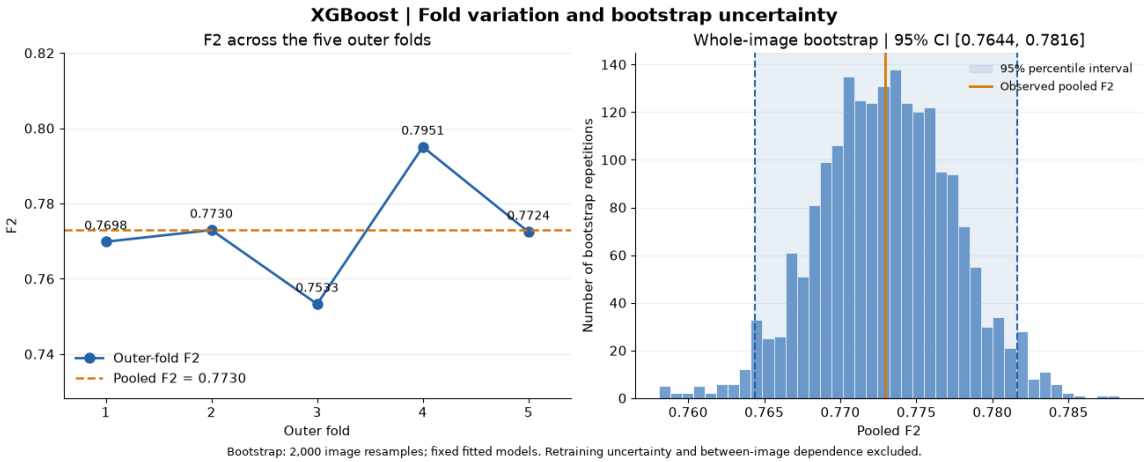


Figure 32. *XGBoost outer evaluation. Left: F2 per outer fold. Right: whole-image bootstrap distribution of the pooled F2 (95% CI 0.764–0.782).*

On the test set XGBoost found 86.0% of the landslide pixels (228,346) and flagged 202,348 background pixels, for F2 = 0.765. Its ranking quality equals that of HGB (ROC-AUC 0.991, average precision 0.773; Figure 33). A diagnostic threshold scan on the test set (Figure 34) shows that F2 is nearly flat between thresholds of about 0.1 and 0.2, so the locked threshold of 0.14, chosen on development data only, lies close to the test optimum. The scan was not used to change the threshold.

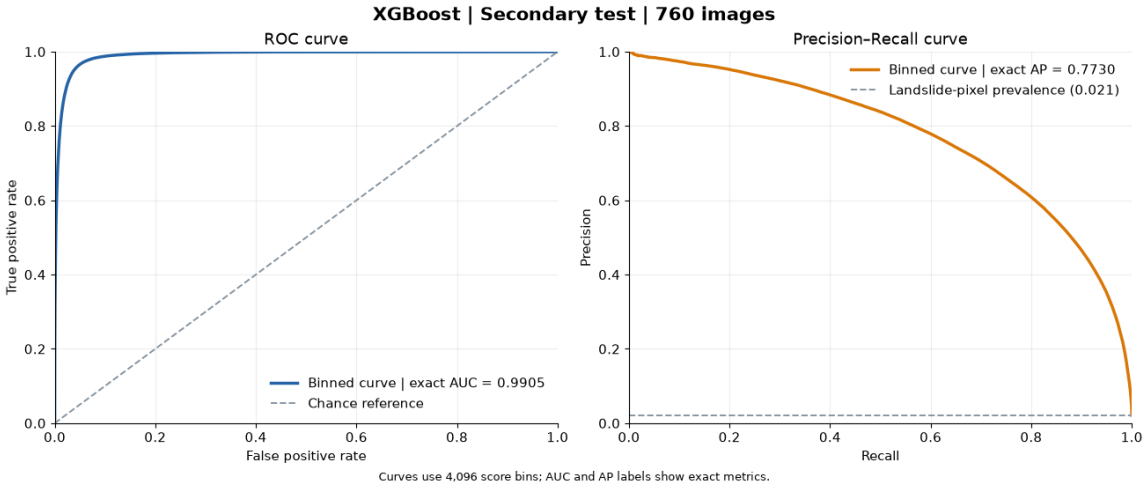


Figure 33. *XGBoost ROC and precision–recall curves on the 760 test images.*

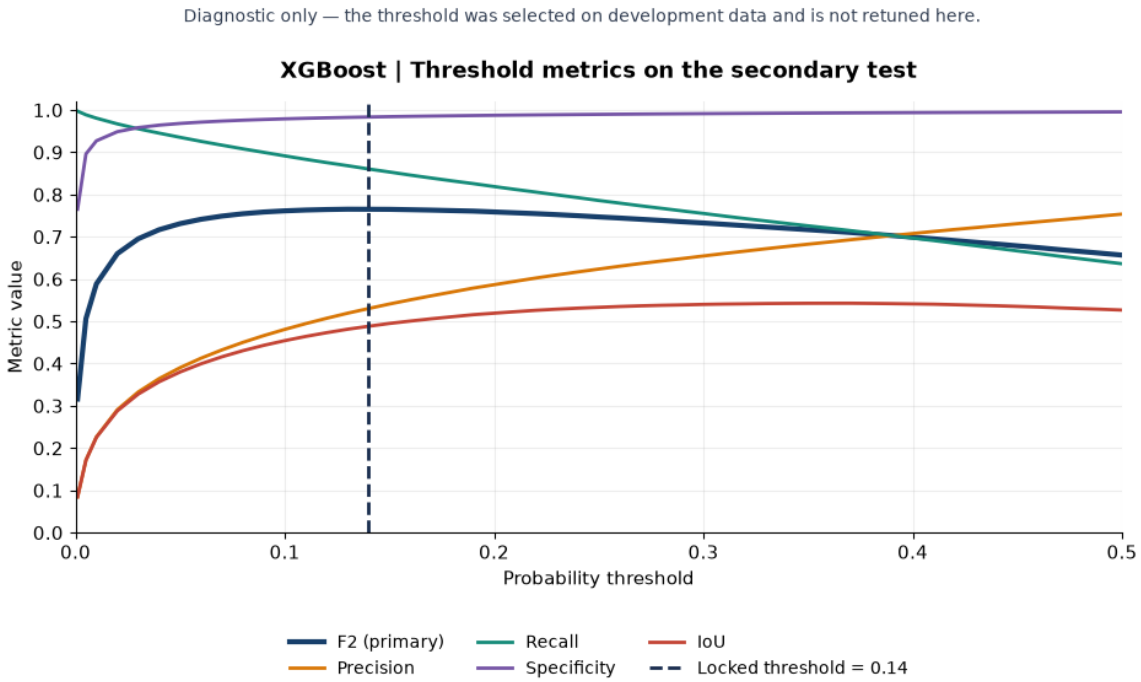


Figure 34. *F2, precision, recall, specificity and IoU of XGBoost across thresholds on the test set (diagnostic only). Dashed line = locked threshold 0.14.*

The six diagnostic images (Figure 35) and the permutation importance (Figure 36) match the HGB results closely: B4 is by far the most important channel (F2 drop 0.44), followed by B1, elevation and slope. XGBoost was also the cheapest ensemble to build: the nested search took about 73 minutes and the final fit 74 seconds.

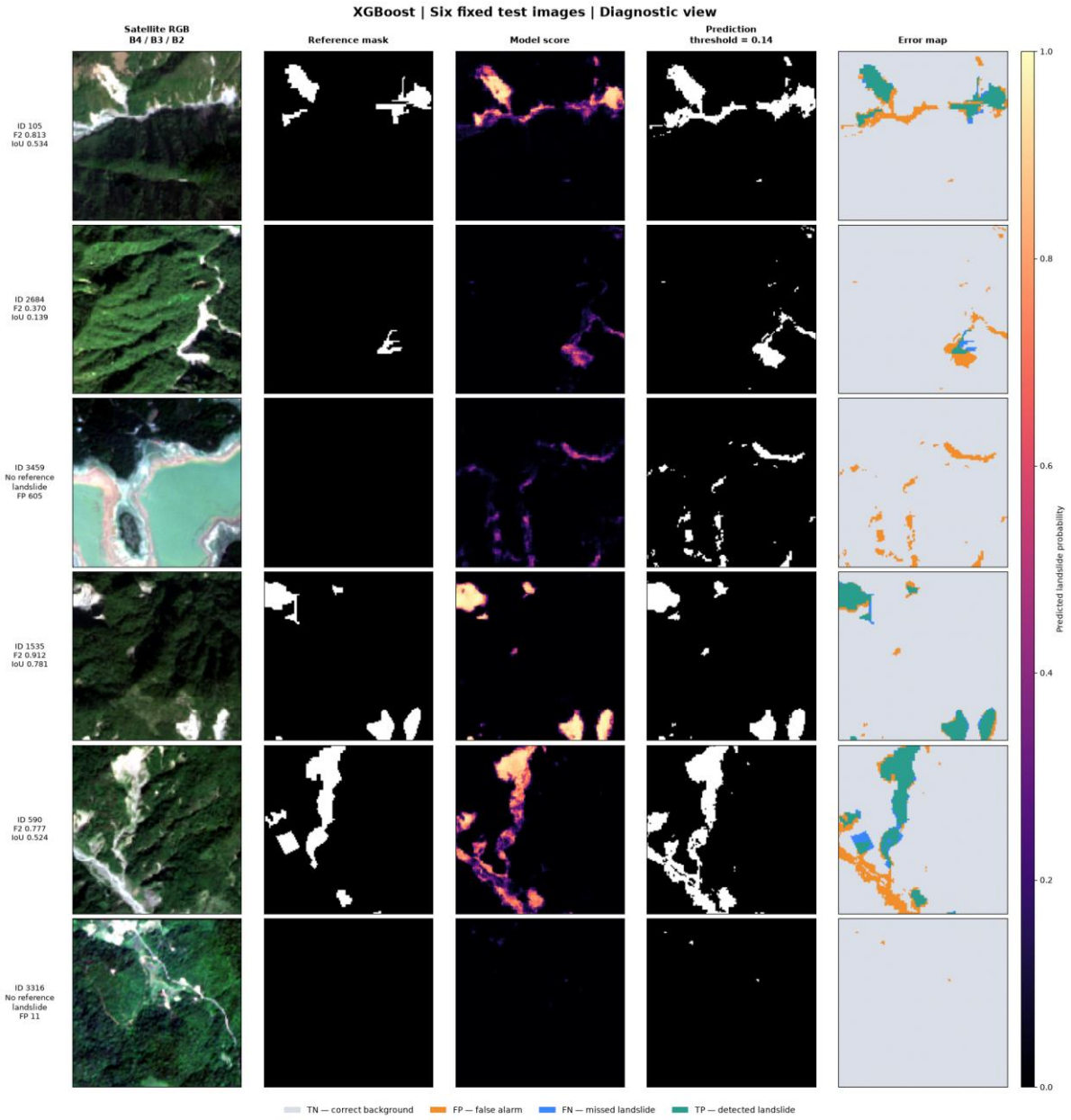


Figure 35. *XGBoost on six fixed test images. Columns: true-colour image, reference mask, predicted probability, prediction, error map.*

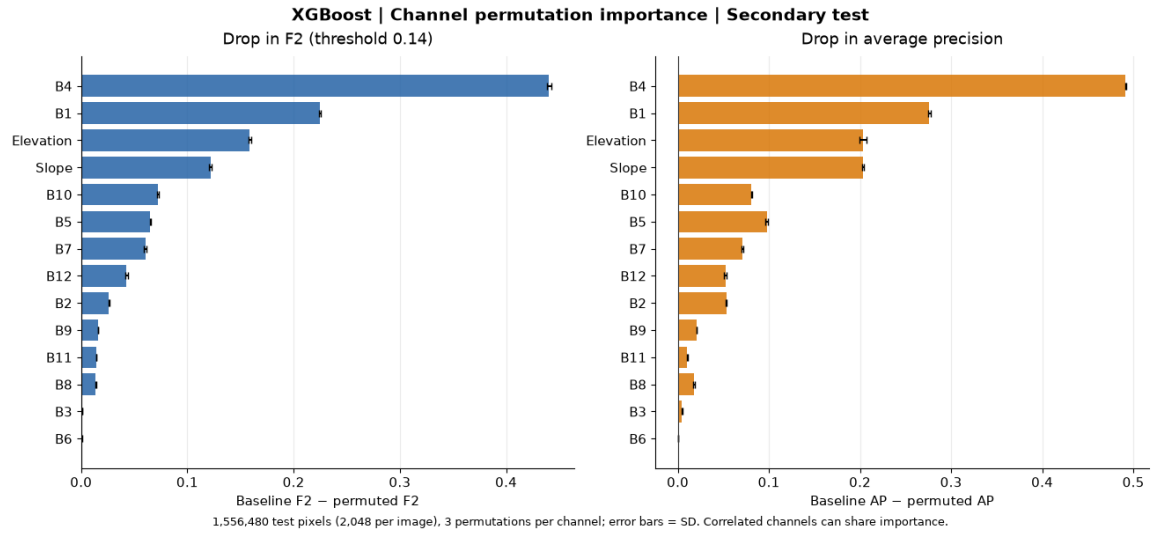


Figure 36. *XGBoost channel permutation importance on the test set. Left: drop in F2. Right: drop in average precision.*

4.2 Overall performance

The random forest, HGB and XGBoost achieved the highest F2 in both evaluations, about 0.77 on the development images and 0.765–0.767 on the test images.

Table 3. *Overall performance on the outer evaluation and the secondary test set.*

Classifier	Outer F2 [95% CI] (3,039 images)	Test F2 [95% CI] (760 images)	Precision	Recall	IoU	Average precision
Random forest	**0.776** [0.768, 0.784]	**0.767** [0.747, 0.786]	**0.535**	0.860	**0.492**	**0.780**
HGB	0.774 [0.765, 0.782]	0.766 [0.747, 0.785]	0.527	**0.864**	0.487	0.774
XGBoost	0.773 [0.764, 0.781]	0.765 [0.745, 0.784]	0.530	0.860	0.488	0.773
LR + interactions	0.748 [0.738, 0.757]	0.743 [0.720, 0.765]	0.511	0.838	0.465	0.738
Decision tree	0.738 [0.728, 0.747]	0.723 [0.700, 0.744]	0.529	0.796	0.466	0.697
LR (raw features)	0.629 [0.614, 0.643]	0.612 [0.577, 0.643]	0.360	0.741	0.320	0.504

**Note: Precision, recall, IoU and average precision refer to the secondary test set. Confidence intervals come from 5,000 paired whole-image bootstrap resamples.*

All classifiers reach high recall but moderate precision. Even the best models find about 86% of landslide pixels, while about every second pixel they mark as landslide is a false alarm. This trade-off is intended by the recall-weighted F2 metric.

Adding the 91 pairwise interaction terms raised logistic regression by +0.119 F2 (outer) and +0.131 F2 (test), three times the gap between the decision tree and the best ensemble. The relationship between the inputs and landslide occurrence is therefore strongly non-linear, which tree-based models capture automatically.

Test F2 was slightly lower than outer F2 for every classifier (−0.005 to −0.018). None of these changes is significant (bootstrap, Bonferroni-adjusted p = 1.0), and their similar size across models points to a slightly harder test-image composition rather than overfitting.

4.3 Statistical comparison

The classifiers form four clearly separated performance groups, and within the leading group the differences are too small to matter. Every difference between groups is significant in both evaluations (paired bootstrap, Bonferroni-adjusted p < 0.001). The Wilcoxon test confirms each comparison with the random forest, with large effect sizes against the random forest (rank-biserial r = −0.86 to −0.90 for the decision tree and raw-feature LR, r = −0.33 for interaction LR). Interaction LR is in turn significantly better than the decision tree (+0.020 F2 on the test set).

Within the ensemble group, the test differences to the random forest were −0.0005 F2 for HGB (95% CI [−0.0030, +0.0023]) and −0.0017 for XGBoost ([−0.0050, +0.0017]), both clearly non-significant (adjusted p = 1.0). On the larger outer evaluation, the random forest was significantly better than HGB (+0.0023, p = 0.006) and XGBoost (+0.0032, p < 0.001). These differences are below 0.5% of the F2 value, a textbook case of statistical significance without practical relevance in very large samples.

The Bayesian analysis resolves this tension. The posterior probability that HGB and XGBoost differ from the random forest by less than 0.01 F2 is 1.00 in both evaluations, as is the probability that HGB and XGBoost are equivalent to each other. On the test set, HGB is better than the random forest in 34% of posterior draws and XGBoost in 16%, so neither direction is favoured. Interaction LR, the decision tree and raw-feature LR are practically worse than the random forest with probability 1.00.

The per-image Wilcoxon test adds one nuance: on typical individual images the random forest is slightly ahead of HGB (r ≈ −0.19, adjusted p < 0.01 in both evaluations), although their pooled pixel-level F2 is equivalent.

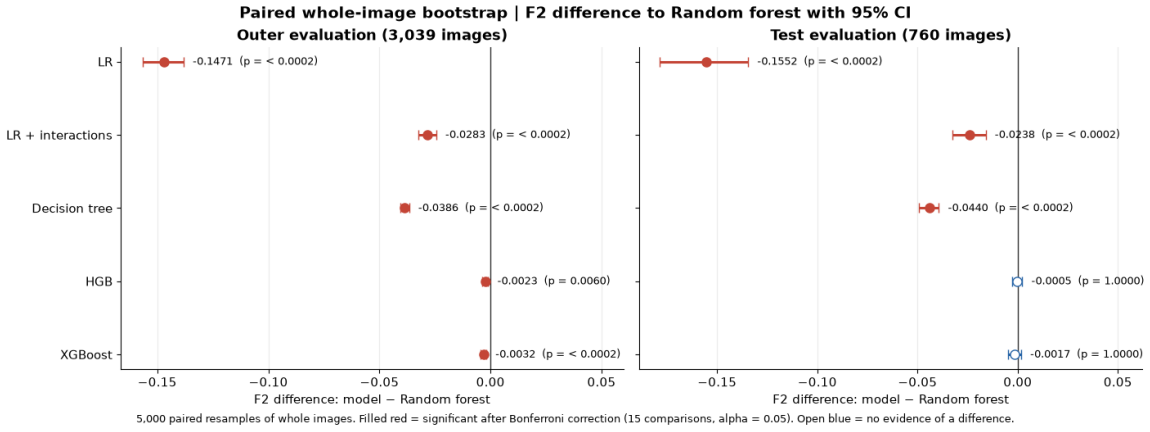


Figure 37. *Paired whole-image bootstrap of the F2 difference between each classifier and the random forest, with 95% confidence intervals. Left: outer evaluation (3,039 images). Right: test evaluation (760 images). Filled red = significant after Bonferroni correction (15 comparisons); open blue = no evidence of a difference.*

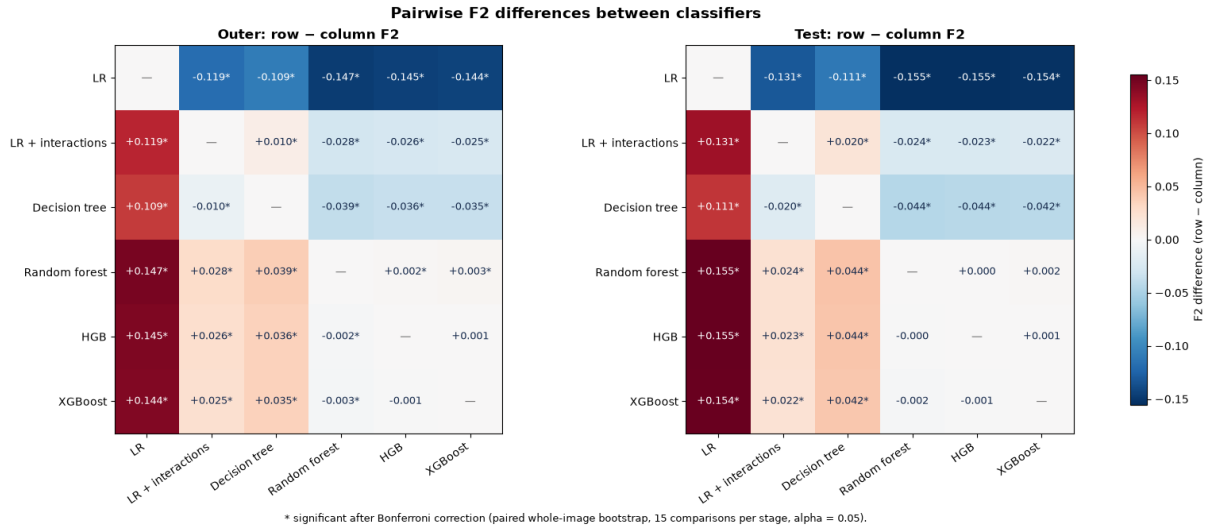


Figure 38. *All pairwise F2 differences (row − column) on the outer (left) and test (right) evaluation. * = significant after Bonferroni correction. The three ensembles form one block with near-zero differences.*

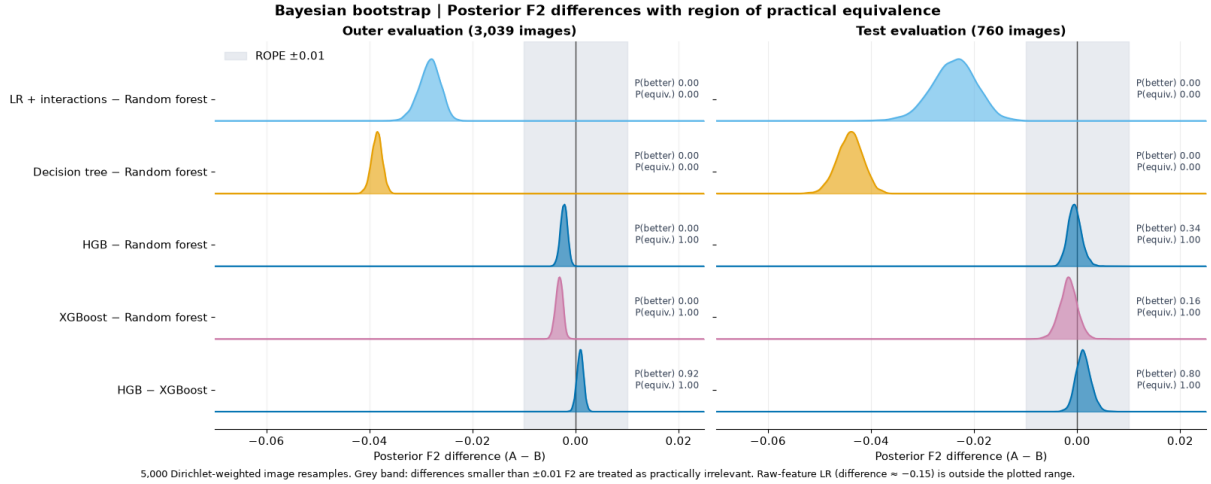


Figure 39. *Bayesian bootstrap posteriors of the F2 difference for five key pairs. Grey band = region of practical equivalence (±0.01 F2). The posteriors of the three ensemble pairs lie entirely inside the band; those of LR + interactions and the decision tree lie entirely outside.*

4.4 Landslide size and false alarms

Detection quality depends far more on landslide size than on the choice of classifier. On the outer evaluation, pooled F2 of the best models rises from about 0.52 in images with at most 1% landslide coverage (610 images) to about 0.88 in images with more than 20% coverage (37 images).

The gaps between classifiers are widest for small landslides. A difference-in-differences test showed that the random forest's advantage over raw-feature LR, the decision tree and interaction LR is about twice as large in images with ≤ 1% coverage as in images with > 5% coverage (outer evaluation, adjusted $p \leq 0.024$). The three ensembles perform equally across all sizes. On the test set the same pattern appears but is significant only for raw-feature LR, because each size group contains far fewer images.

On the 1,254 development and 314 test images without landslides, raw-feature LR marked about 2.0–2.4% of all pixels as landslide, five to seven times more than the ensembles. HGB had the lowest false-alarm rate (0.36% outer, 0.31% test), followed by the random forest (0.39%, 0.34%), XGBoost (0.42%, 0.39%) and the decision tree (0.55%, 0.43%).

Image-level outcomes reveal a small trade-off between the ensembles:

- **Sensitivity:** the random forest detected at least half of the landslide pixels in 91.8% (outer) and 91.5% (test) of landslide images, compared with 90.6–91.0% for HGB and XGBoost.
- **Clean negative images:** HGB and XGBoost left 32.7–35.4% of landslide-free images completely without false alarms, compared with 28.9–32.5% for the random forest.

Both differences are significant on the outer evaluation (exact McNemar, adjusted $p \leq 0.011$) but not on the smaller test set. The median per-image F2 on test images with landslides was 0.714 for the random forest, 0.711 for XGBoost, 0.701 for HGB, 0.695 for interaction LR, 0.651 for the decision tree and 0.576 for raw-feature LR.

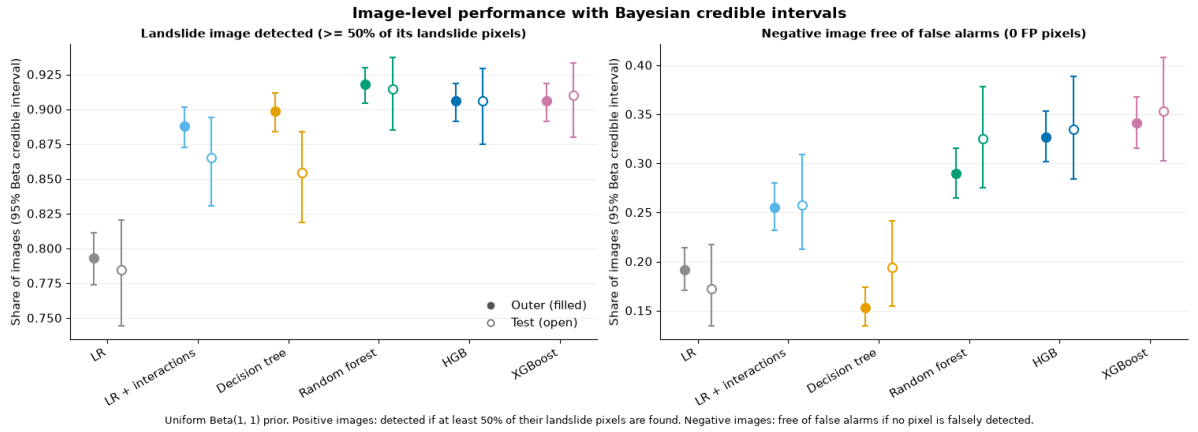


Figure 40. *Pooled F2 of each classifier by landslide coverage of the image, for the outer (left) and test (centre) evaluation with 95% bootstrap intervals, and the share of falsely flagged pixels on images without landslides (right; solid = outer, hatched = test).*

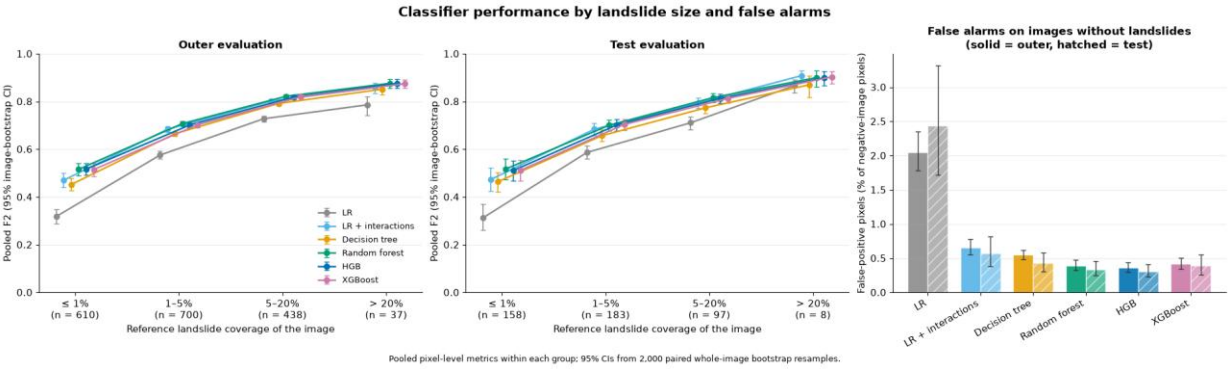


Figure 41. *Distribution of per-image F2 on images with landslides (outer: 1,785; test: 446). Each row shows the density, a box plot (median and interquartile range) and the individual images. All classifiers have long lower tails: a sizeable share of images, mostly with tiny landslides, is detected poorly by every model.*

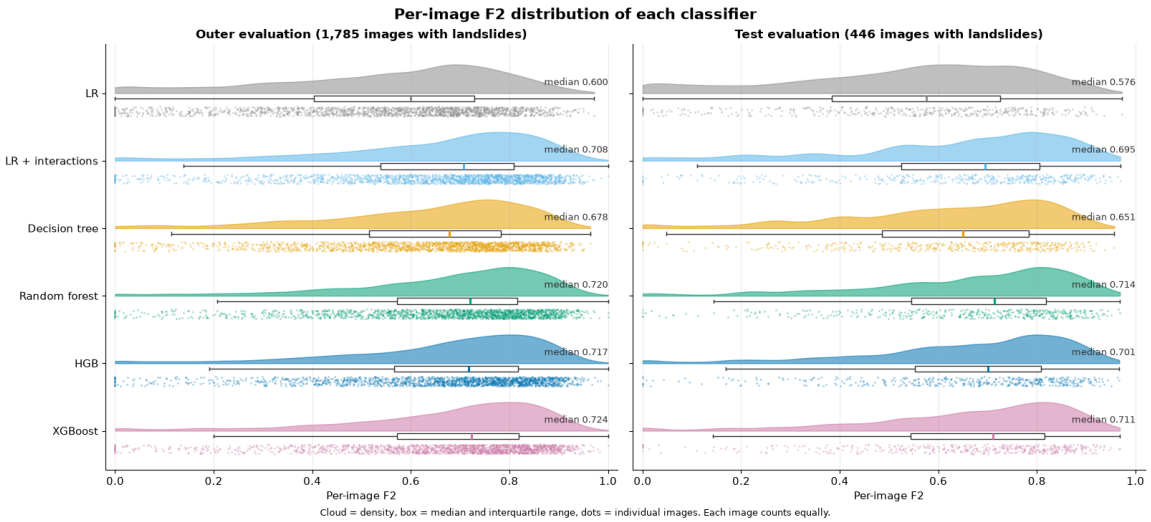


Figure 42. *Image-level outcomes with 95% Bayesian credible intervals (Beta(1,1) prior). Left: share of landslide images in which at least 50% of the landslide pixels were detected. Right: share of landslide-free images without a single false-positive pixel. Filled = outer, open = test.*

4.5 Ranking quality and shared errors

The three ensembles rank pixels almost identically: their precision–recall curves overlap over the full recall range, with average precision of 0.780 (random forest), 0.774 (HGB) and 0.773 (XGBoost). Interaction LR follows with 0.738, the decision tree with 0.697 and raw-feature LR with 0.504; the latter never exceeds a precision of about 0.75, even for its most confident predictions. Each classifier's locked operating point lies close to where its curve touches the highest reachable F2 iso-line, confirming that thresholds chosen on development data are near-optimal on new images.

The classifiers largely agree on what is easy and what is hard. Of all 265,381 landslide pixels in the test set, 66.5% were detected by all six classifiers and 8.0% were missed by all of them; among pixels missed by at least one classifier, one in four (24%) was missed by every classifier. Of the non-landslide pixels flagged by at least one classifier, 14.9% (77,278 pixels) were flagged by all six.

The consensus maps of six fixed test images show where these shared errors occur:

- **Compact, bright scars** (images 1535 and 590) are detected by all classifiers.
- **River channels and debris-covered valley floors** next to real landslides (images 105 and 590) are flagged by most or all classifiers.
- **Small landslides** (image 2684, 0.7% coverage) are only partly found, while a nearby bright stream is flagged.
- **Reservoir water and banks** (image 3459, no landslide) are flagged widely by a single classifier, consistent with the many false alarms of raw-feature LR; the ensembles raise only scattered alarms along the banks.
- **Fully vegetated terrain** (image 3316) is almost free of false alarms for all classifiers.

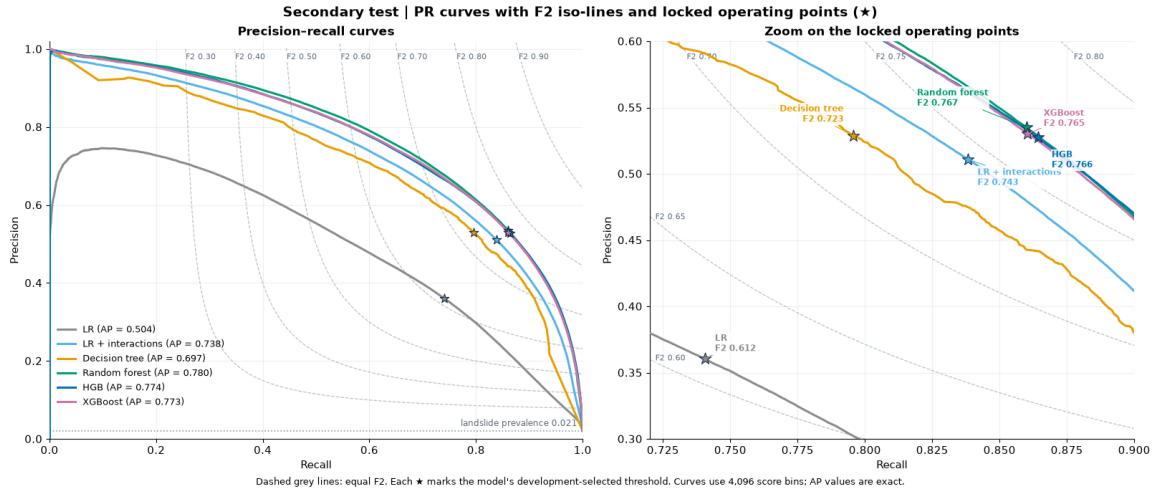


Figure 43. *Precision–recall curves of all classifiers on the test set with F2 iso-lines (dashed grey). Left: full curves. Right: zoom on the locked operating points (diamonds). The three ensemble curves overlap almost completely.*

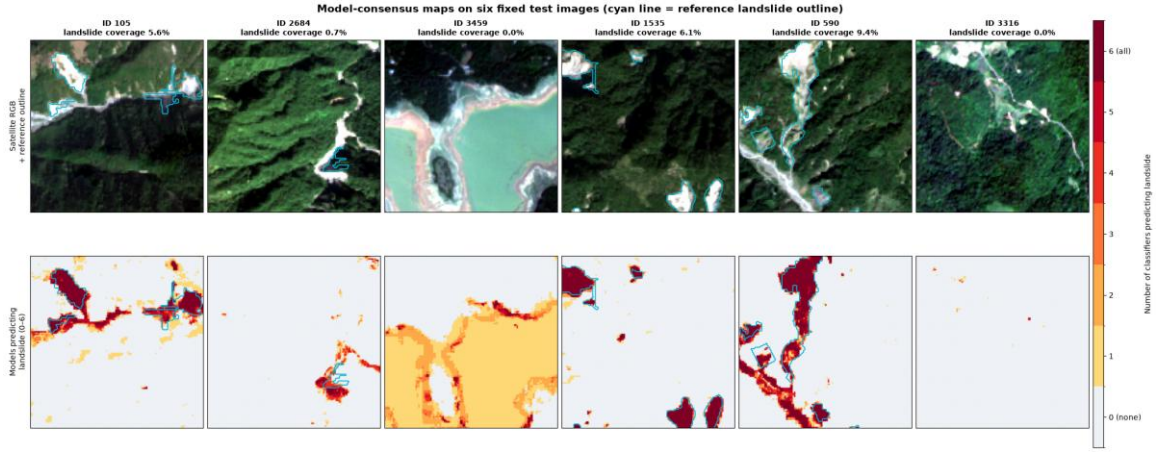


Figure 44. *Model consensus on six fixed test images. Top: true-colour image with the reference outline (cyan). Bottom: number of classifiers (0–6) that flag each pixel. Dark red = all six agree. Compact scars (IDs 1535 and 590) are found by all six models, while the shared false alarms follow bright river channels (IDs 105 and 590). The wide yellow area on the reservoir shore (ID 3459) is flagged by only one or two models.*

4.6 Probability calibration and computing cost

The classifiers differ far more in probability quality and computing cost than in F2. Prediction time was measured for every final model on the same 100 test images (1.64 million pixels, 2 CPU threads, median of 3 runs); every model reproduced its saved test scores exactly.

The random forest produces the best-calibrated probabilities, and its lower Brier score is significant against every other classifier (Holm-corrected $p < 0.001$), although the margin over HGB and XGBoost is only 0.0002. The decision tree is strongly miscalibrated: its few distinct leaf probabilities overstate the landslide probability, which explains its unusually high optimal threshold of 0.92.

Table 4. *Probability calibration and prediction time on the secondary test set.*

Classifier	Brier score	ECE	MCE	Prediction time (s per million pixels)
Random forest	**0.0088**	**0.0010**	**0.015**	4.69
HGB	0.0090	0.0018	0.087	6.53
XGBoost	0.0090	0.0011	0.053	1.85
LR + interactions	0.0098	0.0027	0.092	0.69

Classifier	Brier score	ECE	MCE	Prediction time (s per million pixels)
Decision tree	0.0356	0.0426	0.746	**0.05**
LR (raw features)	0.0140	0.0042	0.228	0.09

**Note: Calibration on the secondary test set, 15 uniform probability bins. ECE and MCE: expected and maximum calibration error.*

Among the three equivalent ensembles, XGBoost predicts 2.5 times faster than the random forest and 3.5 times faster than HGB. Training effort differs even more. Fitting the final model took 27 s for the decision tree, 63 s for HGB and 74 s for XGBoost, compared with roughly 8–11 minutes for each random-forest refit. The complete nested search took about 73 minutes for XGBoost but on the order of 40 hours for the random forest.

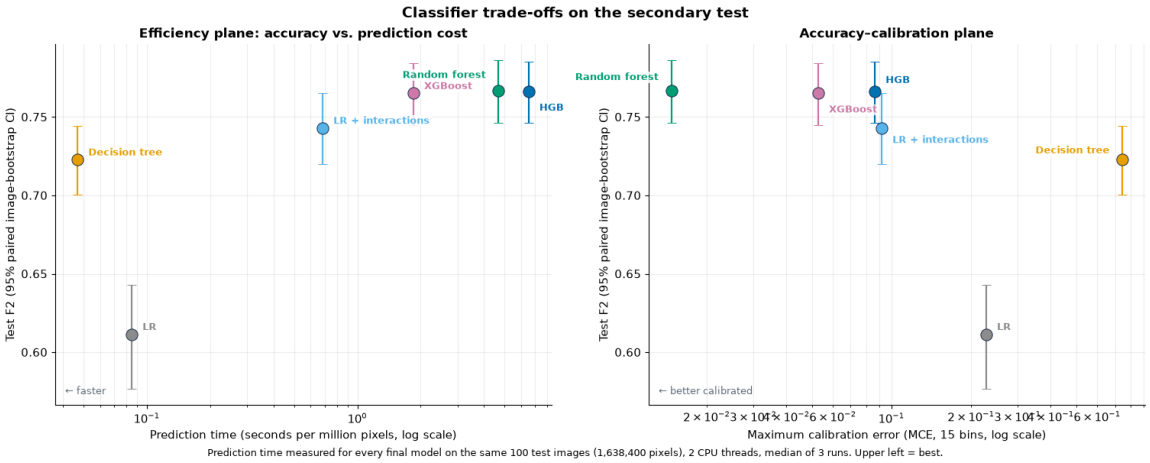


Figure 45. *Trade-offs on the test set. Left: F2 (95% CI) against prediction time per million pixels (log scale; up and left is better). Right: F2 against maximum calibration error (up and left is better).*

4.7 What the classifiers rely on

All tree-based classifiers rely most strongly on the red band B4. Permuting it reduced F2 by 0.41 for HGB and 0.44 for XGBoost, and it ranked first for the random forest and the decision tree as well. Elevation and slope were among the four most important inputs for the random forest, HGB and XGBoost, so terrain adds information beyond the spectral bands. The green band B3 and the red-edge band B6 contributed almost nothing on their own, most likely because neighbouring bands carry the same information. The importance methods differed slightly between notebooks, so only rankings are compared.

The exact SHAP explanation of XGBoost shows how the inputs act:

- **B4 (red) is the main positive evidence.** Its contribution rises steadily with brightness and adds on average +2.1 log-odds to landslide pixels.
- **B1 acts as a rule-out variable.** Low values push strongly towards "no landslide" (down to about −5 log-odds); high values add only a moderate amount.
- **Vegetation-sensitive bands (B7, B8) act as a veto.** High values push against a landslide.
- **Terrain is physically plausible.** Steep slopes raise and flat terrain lowers the landslide probability.

The explanation maps make the main error source visible. On riverbeds and debris-covered valley floors, B4 pushes towards "landslide" just as strongly as on real scars, and the vegetation bands provide no counter-evidence on these bare surfaces. B1 and elevation vary smoothly across whole images (B1 even shows its coarse 60 m pixel grid), which suggests that they partly act as scene- or region-level variables rather than landslide-specific signals.

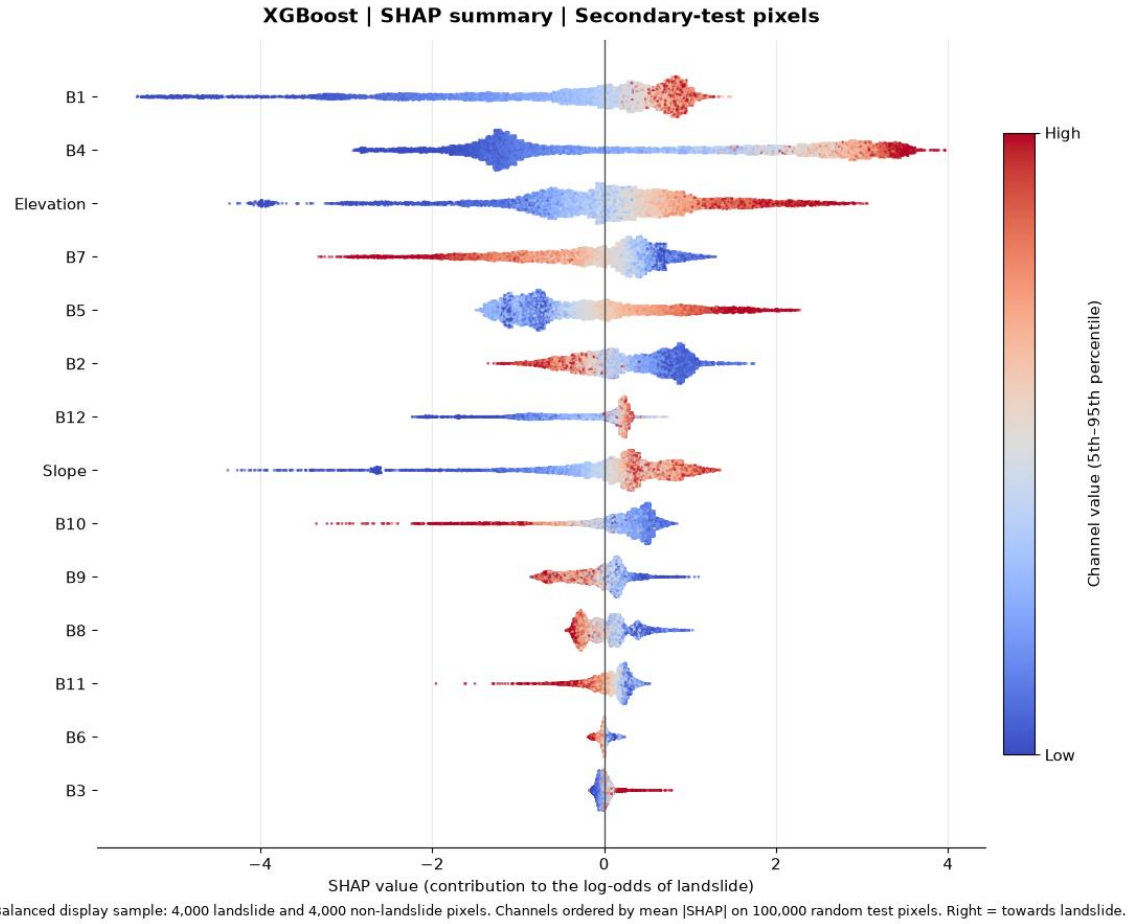


Figure 46. *SHAP summary of the final XGBoost model on test pixels (balanced display sample of 4,000 landslide and 4,000 non-landslide pixels). Each dot is one pixel; colour = channel value (red = high); positive values push the prediction towards landslide. High B4 is the strongest positive evidence, while low B1 strongly argues against a landslide.*

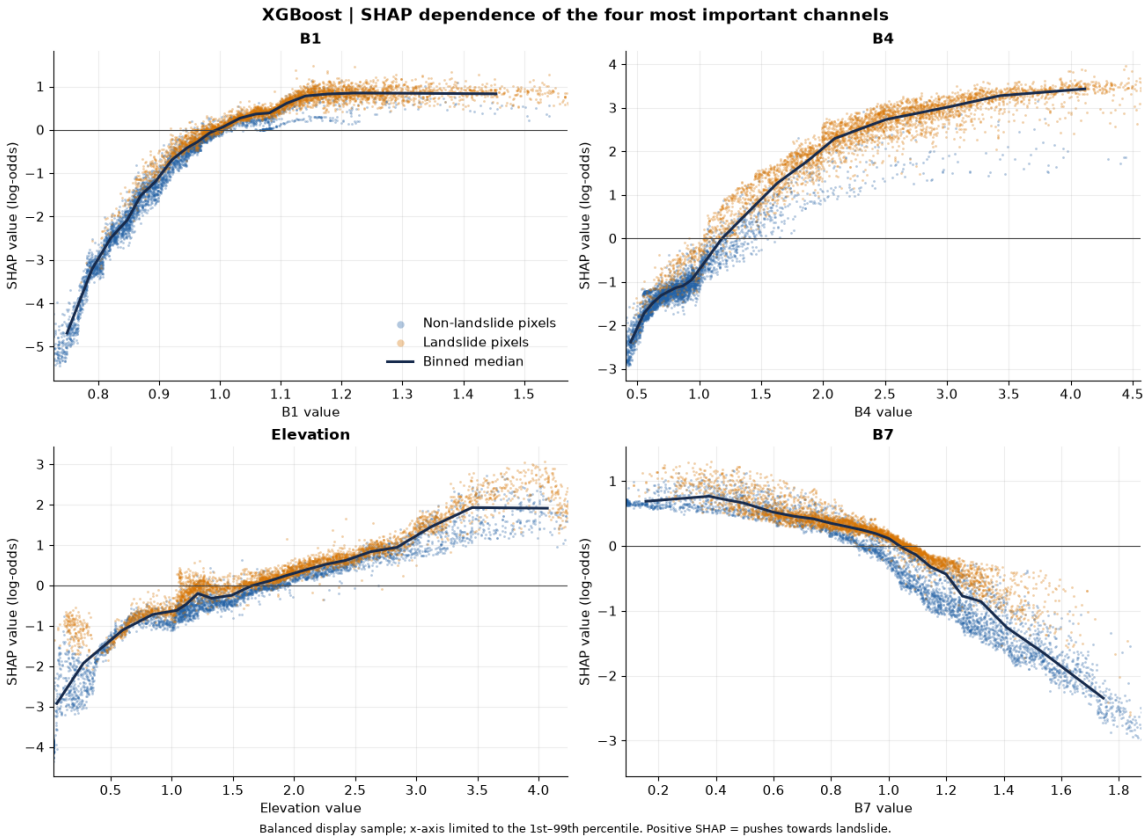


Figure 47. *SHAP dependence of the four most important channels (B1, B4, elevation, B7). Black line = binned median. The effects are smooth and monotonic in the main data range, consistent with physical expectations.*

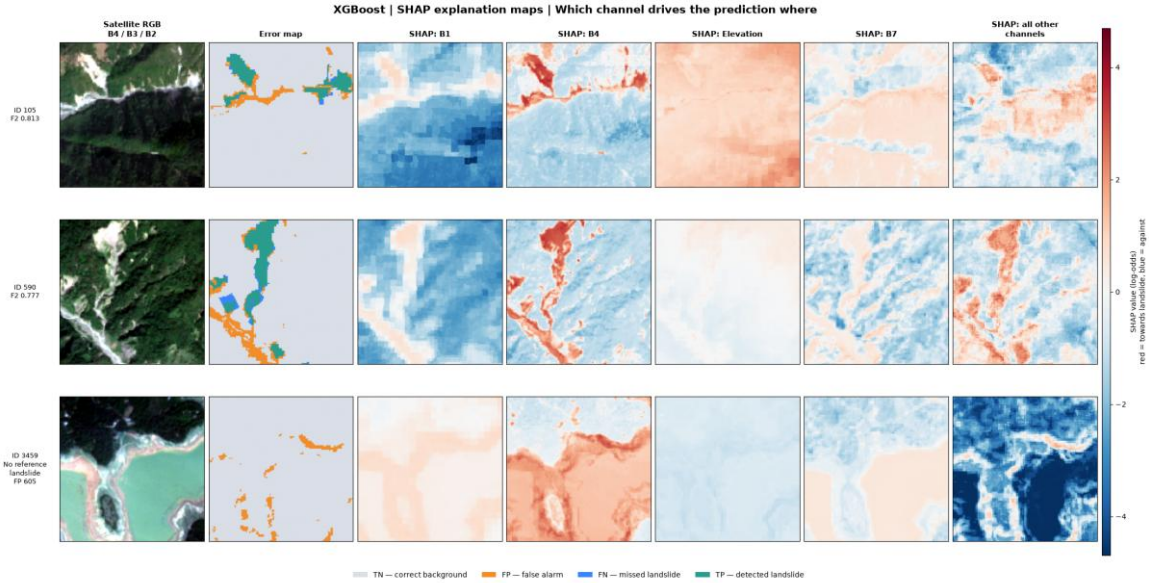


Figure 48. *SHAP explanation maps of three test images. Columns: true-colour image, error map, and the SHAP contribution of B1, B4, elevation, B7 and all other channels (red = towards landslide, blue = against). Along the riverbed in the middle row (ID 590), B4 pushes strongly towards landslide and produces the false alarms. On the reservoir shore in the bottom row (ID 3459), B4 again argues for landslide, but the other channels largely cancel it, so only small false-alarm patches remain.*

5. Discussion

5.1 Why the ensembles lead — and why they plateau

The ranking follows model flexibility. Logistic regression on the raw channels can only draw a straight boundary in feature space, and it misses the combinations that separate landslides from other bright surfaces. Adding explicit pairwise interactions closes most of that gap (+0.13 F2), and a single decision tree finds such combinations automatically but overfits (its training F2 exceeded its outer-evaluation F2 by about 0.05) and produces coarse, poorly calibrated probabilities. The ensembles combine many trees and remove most of this variance.

That three very different ensemble algorithms — bagging (random forest) and two forms of gradient boosting — converge on practically the same F2 is itself an important result. It indicates that, with these 14 inputs, the achievable pixel-level performance is limited by the information each pixel carries rather than by the learning algorithm. The shared errors support this: 8% of landslide pixels are missed by every classifier, and 14.9% of all false alarms are made by every classifier.

5.2 The physical reason for the remaining errors

The errors are not random. SHAP shows that the models rely on a physically sensible rule — bright in the red band, not vegetated, on a steep slope — and this rule cannot tell a fresh landslide scar from a riverbed, a debris-covered valley floor or a bare reservoir bank. A pixel classifier sees one pixel at a time and has no notion of shape, size or surroundings. Very small landslides suffer for the same reason: a few mixed pixels at the edge of a scar carry too little evidence, which is why F2 falls to about 0.52 for images with at most 1% landslide coverage.

These errors point clearly to the next step. A human interpreter separates a landslide from a river by its elongated or fan-shaped form, its position on the slope and its connection to a source area. Methods that see spatial context can exploit exactly this information.

5.3 Statistical versus practical significance

The comparison illustrates why significance tests alone can mislead. On the 3,039 development images, the random forest beat HGB and XGBoost "significantly", yet by only 0.002–0.003 F2; on the 760 test images the same differences were not significant. Neither result alone settles the question, because "not significant" does not mean "equal". The Bayesian analysis with a pre-defined region of practical equivalence gives a direct answer: the three ensembles are practically equivalent with posterior probability 1.00. Treating images as the unit of analysis was essential throughout: tests on 12 million dependent pixels would have declared almost any difference significant.

5.4 Which classifier to use

Because the ensembles are equivalent in F2, the choice depends on the application:

- **XGBoost** is the most practical choice for large-area or rapid-response mapping: it predicts 2.5–3.5 times faster than the other ensembles, is well calibrated (MCE 0.053), needs a small fraction of the random forest's training effort, and leaves the most landslide-free images completely clean.
- **Random forest** suits applications that use probabilities directly, such as landslide-probability maps, because it is the best calibrated model (MCE 0.015). It is also slightly more sensitive, finding a few more landslide images, which matters when missed landslides are most costly.
- **Interaction LR** is a transparent and fast alternative when interpretability or minimal computing cost matters, at a cost of about 0.02–0.03 F2.

The single decision tree and raw-feature LR are not recommended: they are clearly less accurate, and the decision tree's probabilities cannot be interpreted as probabilities.

6. Limitations

The results describe transfer to new images from the same four study areas, not to new regions. The main limitations are:

1. **No geographic independence.** Development and test patches come from the same regions and events, and the files carry no coordinates. Neighbouring patches may share terrain and illumination, so the test results are likely somewhat optimistic for a new region or a new event.
2. **Reused test set.** The test images were first examined during the logistic regression analysis. No model was tuned on them, but they are reported as a secondary rather than a fully untouched test.
3. **Uncertainty is conditional.** All bootstrap intervals keep the fitted models fixed and treat images as independent. They do not include the variability of retraining or the spatial dependence between neighbouring patches.
4. **Unequal and compact searches.** The number of tuned configurations ranged from 3 (XGBoost) to 25 (decision tree), and the boosting grids were small, with the selected settings at the edge of each grid. Slightly better settings may exist, although all candidates differed by less than 0.014 F2.
5. **Reference data.** Some reference masks contain simplified geometric shapes, which lowers the measured precision and IoU even where a model follows the visible scar.
6. **Scaled inputs.** Slope and elevation are stored as scaled values and could not be interpreted in physical units.
7. **Single machine.** Computing times were measured on one laptop; their ratios are more reliable than their absolute values.
8. **Choice of equivalence margin.** The ±0.01 F2 equivalence margin is a judgement. It was fixed in advance and is smaller than the natural fold-to-fold variation, but a stricter margin could change the verdict for the smallest differences.

7. Conclusion and future work

7.1 Conclusion

Pixel-based supervised classification can map landslides from Sentinel-2 and terrain data with high recall: the best classifiers find about 86% of landslide pixels on held-out images (F2 ≈ 0.77). Random forest, HGB and XGBoost are practically equivalent, clearly ahead of interaction logistic regression, a single decision tree and raw-feature logistic regression. Non-linear feature interactions are the key ingredient, worth about 0.13 F2 for logistic regression.

The remaining errors are shared by all classifiers and have a clear physical cause: bright, bare surfaces that resemble fresh landslide scars, and very small landslides with too little evidence per pixel. Further tuning of pixel-based classifiers is therefore unlikely to bring large gains. In practice, XGBoost offers the best balance of accuracy, calibration and computing cost, while the random forest provides the most reliable probabilities.

Methodologically, the project shows that a fair comparison needs more than a results table: identical image-level splits, thresholds chosen on development data only, tests that respect the dependence between pixels, and a Bayesian equivalence analysis that can confirm when differences do not matter.

7.2 Future work

The clearest next step is to give the models spatial context, beyond the methods covered in this course.

- **Deep learning for semantic segmentation.** Convolutional networks such as U-Net and transformer-based networks such as SegFormer and Swin-Transformer encoders classify each pixel using its whole neighbourhood and learn shape and texture directly. The leading entries of the Landslide4Sense 2022 competition used exactly such networks, combined with self-training and post-processing (Ghorbanzadeh et al., 2022b).
- **Using the unlabelled patches.** The 1,045 unlabelled validation and test patches of the dataset could be used for self-training (pseudo-labelling) or active learning.
- **Geospatial foundation models.** Models pre-trained on large amounts of satellite imagery, such as Prithvi from IBM and NASA, can be fine-tuned with few labels and may transfer better to new regions.
- **Spatial context for classical models.** Neighbourhood statistics, texture measures, spectral indices such as NDVI, or object-based image analysis would let the existing classifiers see beyond one pixel at low cost. Interpretable models such as explainable boosting machines would keep the results transparent.
- **Post-processing.** Morphological filtering or conditional random fields could remove isolated false alarms and smooth landslide outlines.
- **Change detection and additional sensors.** Pre- and post-event image pairs would separate new landslides from permanently bare surfaces such as riverbeds, the main source of false alarms. Sentinel-1 radar data would add information that is available through clouds, which often cover an area right after heavy rainfall.
- **Stricter validation.** Leave-one-region-out or spatially blocked cross-validation would show how well models transfer to unseen regions and events.

7.3 Further applications

With these extensions, the approach could support:

- **Rapid-response mapping** after earthquakes, typhoons and extreme rainfall, delivering first landslide maps within days of a cloud-free acquisition.
- **Impact assessment** by overlaying detected landslides with roads, buildings, power lines and rivers to prioritise rescue and repair.
- **Landslide inventories** for large regions, which are the basis for susceptibility and hazard maps and for land-use planning.
- **Probabilistic risk maps** that use calibrated probabilities, such as those of the random forest, together with uncertainty estimates instead of a single yes/no map.
- **Multi-hazard monitoring** that combines landslide detection with flood and building-damage mapping from the same satellite data.

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Data sources: Landslide4Sense on Kaggle · Landslide4Sense dataset documentation

Appendix

A. Size dependence of the random-forest advantage

Difference-in-differences = (random forest − classifier) in images with ≤ 1% landslide coverage minus the same difference in images with > 5% coverage; positive values mean the random forest's advantage is larger for small landslides. Outer evaluation: 610 small and 475 large images; test: 158 and 105.

Table 5. *Size dependence of the random-forest advantage (difference-in-differences).*

Classifier	Outer: advantage small / large	Outer: difference [95% CI]	Outer: adjusted p	Test: difference [95% CI]	Test: adjusted p
LR (raw features)	+0.198 / +0.093	+0.104 [+0.074, +0.132]	< 0.001	+0.112 [+0.052, +0.167]	< 0.001
Decision tree	+0.064 / +0.029	+0.035 [+0.025, +0.044]	< 0.001	+0.012 [−0.002, +0.028]	0.49
LR + interactions	+0.045 / +0.020	+0.026 [+0.008, +0.044]	0.024	+0.028 [+0.003, +0.055]	0.11
HGB	−0.001 / +0.003	−0.004 [−0.013, +0.005]	1.00	+0.004 [−0.008, +0.015]	1.00
XGBoost	+0.001 / +0.002	−0.001 [−0.009, +0.007]	1.00	+0.003 [−0.007, +0.012]	1.00

B. Transfer from the outer to the test evaluation

Table 6. *Change in F2 from the outer to the secondary test evaluation.*

Classifier	Change in F2 (test − outer)	95% CI
Random forest	−0.009	[−0.031, +0.011]
HGB	−0.008	[−0.029, +0.013]
XGBoost	−0.008	[−0.030, +0.013]
LR + interactions	−0.005	[−0.030, +0.020]
Decision tree	−0.015	[−0.039, +0.008]
LR (raw features)	−0.018	[−0.054, +0.017]

No change is significant (Bonferroni-adjusted p = 1.00 for all). With 760 test images, decreases of up to about 0.03 F2 cannot be ruled out.

C. Image-level outcomes

Table 7. *Image-level outcomes of each classifier.*

Classifier	Landslide images detected (outer / test)	Landslide-free images without any false alarm (outer / test)
Random forest	**91.8% / 91.5%**	28.9% / 32.5%
HGB	90.6% / 90.6%	32.7% / 33.4%
XGBoost	90.6% / 91.0%	**34.1% / 35.4%**
LR + interactions	88.8% / 86.5%	25.5% / 25.8%
Decision tree	89.9% / 85.4%	15.3% / 19.4%
LR (raw features)	79.3% / 78.5%	19.1% / 17.2%

A landslide image counts as detected when at least 50% of its landslide pixels are found. Cochran's Q rejects equal rates across the six classifiers for both outcomes and both evaluations ($p < 10^{-25}$).

D. Secondary metrics among the ensembles

On the outer evaluation, the random forest had significantly higher precision than HGB (+0.011) and XGBoost (+0.008), and higher specificity and IoU, while HGB had significantly higher recall (+0.003; test +0.004). On the test set, the precision and IoU differences between the ensembles were not significant (Holm-corrected).

E. Normality check of the per-image differences

The per-image F2 differences to the random forest are clearly not normally distributed: the points in the Q-Q plots bend away from the reference line in both tails, and Shapiro–Wilk tests reject normality for every pair ($p < 10^{-15}$). This is why the comparison relies on the bootstrap and the Wilcoxon signed-rank test instead of a paired t-test.

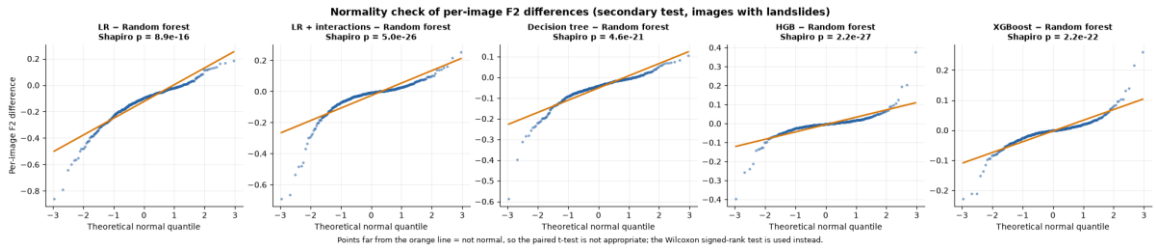


Figure 49. *Normal Q-Q plots of the per-image F2 differences (classifier − random forest) on the 446 test images with landslides.*